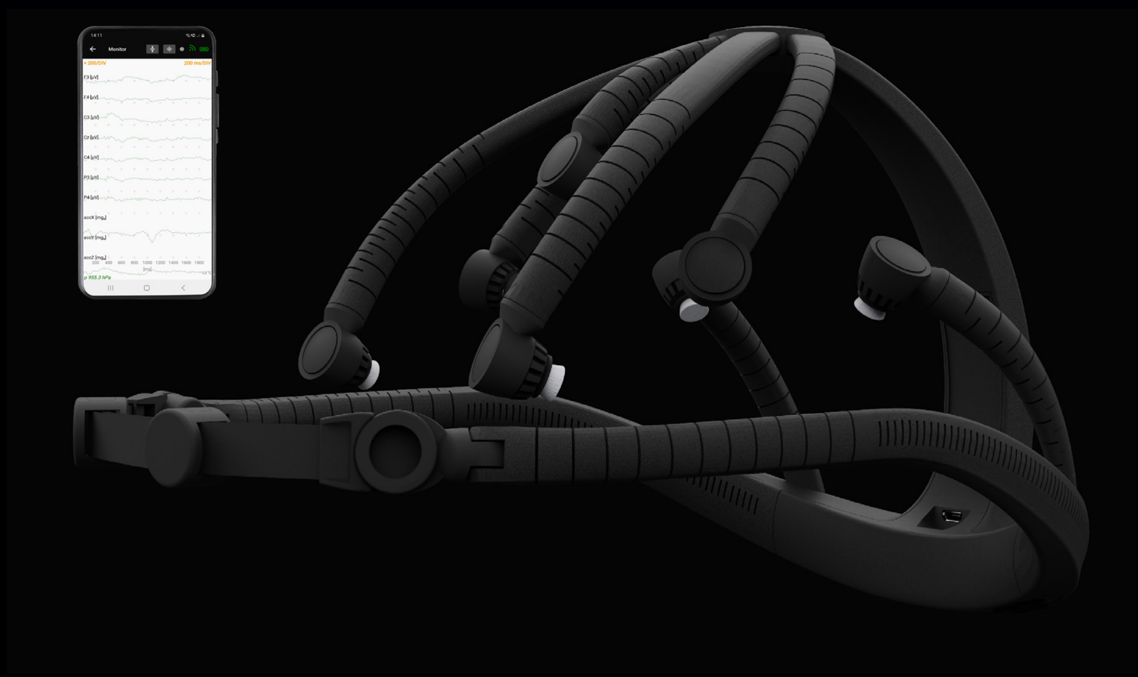
USER MANUAL



EEG Headset rev. 01 and 02

X.on App for Android™ v1.1

X.on App for PC v1.0

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For the latest version of this document, please visit www.xon-eeg.com/downloads or contact Brain Products at info@xon-eeg.com.

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1 Safety Information

Please read the following safety information carefully since it helps to prevent personal injury and damage to property. It is assumed that you have the required specialist knowledge in handling the product and accessories.

Brain Products will not accept any liability for loss or damage resulting from a failure to follow these operating instructions and, in particular, the safety instructions.

1.1 Intended Use

X.on is intended to be used by trained individuals for acquiring neuro-/electrophysiological signals (e.g., EEG, EMG, EOG).

The product may be used by professionals in the context of non-medical applications.

X.on is not a medical device. The use for medical diagnosis, medical therapy, medical monitoring of vital physiological processes (such as cardiovascular functions, etc.) or other medical purposes is expressly forbidden.

1.2 Correct Use

X.on is permitted to be used:

- by professionals in the context of non-medical applications.
- in research institutes and other non-medical environments.

X.on must not be used:

- by laymen or private users.
- by persons who cannot read (e.g., due to visual impairment) or understand (e.g., due to a lack of language skills) this document.
- in MR scanner environment.
- in vicinity of explosive gases, for example in operating theaters.
- in oxygen enriched atmospheres.
- under water (e.g., sea, swimming pool, bath tub) or in environments in which significant amounts of water could enter the products (e.g., under shower, under water tap).

The user is solely liable for any risks to the participants wearing the headset associated with the investigation if the product is not used in accordance with the correct use.

1.3 General Precautions

- Do not open the products and do not modify.
- Never remove the batteries from their housing. Doing so can cause a short circuit and lead to overheating of the batteries.
- Handle the product and its accessories with care.
- No liquids must penetrate the products.
- Do not drop the product or allow it to fall and avoid impacts.
- Heat, direct sunlight (UV radiation), moisture, dust, liquids, conductive foreign matter and excessive radiation can shorten the lifetime of the product.
- Use the supplied cables. Brain Products is not liable for damage caused by cables that are not supplied by Brain Products.

1.4 Risk of Damage to Property

Risk of damage to X.on in the MR scanner room

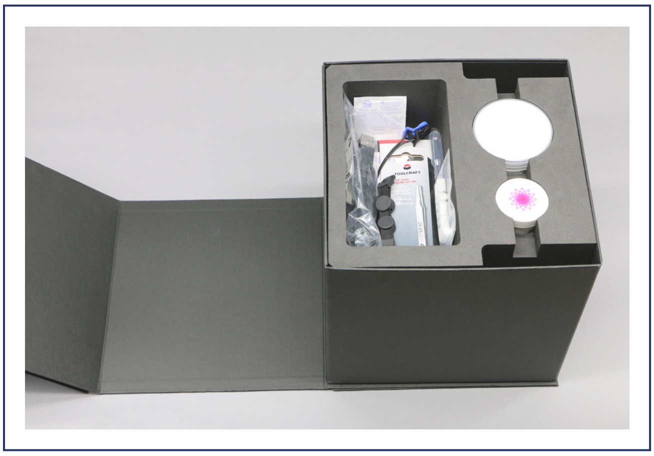
The product is neither MR safe nor MR conditional. It contains magnetic materials. X.on can become a projectile by the magnetic force of the MR scanner. Do not use it in the MR scanner room. You can use the product in the control room of an MR scanner.

1.5 Note on Data Acquisition

During operation, keep a distance of at least 1 m from cell phones or similar transmitters, other sources of high-level interference such as microwave, x-ray or similar equipment that may cause sparks. Otherwise, information may be lost or corrupted.

2 What's in the Case?

Your X.on will arrive in a cube-shaped case with two layers. Take out the upper layer, including all of the additional components, to reveal the X.on headset:



In the case you will find the following components:

1	X.on headset
2	Straps (two lengths; long strap attached to the headset)
3	Ear clip (GND/REF)
4	Jar with sponges and holders
5	Jar with NaCl (including spoon)
6	Pair of tweezers
7	Pipette
8	USB charging cable
9	Extra sponges
10	Alcohol wipes

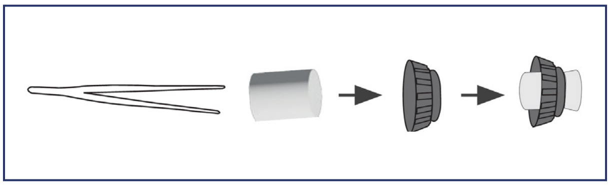
3 Using X.on

Familiarize yourself with the operation of X.on and carry out some test measurements before you start acquiring signals.

3.1 Preparing the X.on for Use

For best signal and data quality, only use the X.on on test participants with freshly washed and dry hair without any further hair treatments or styling products and make sure that the sponges are in direct contact with the skin.

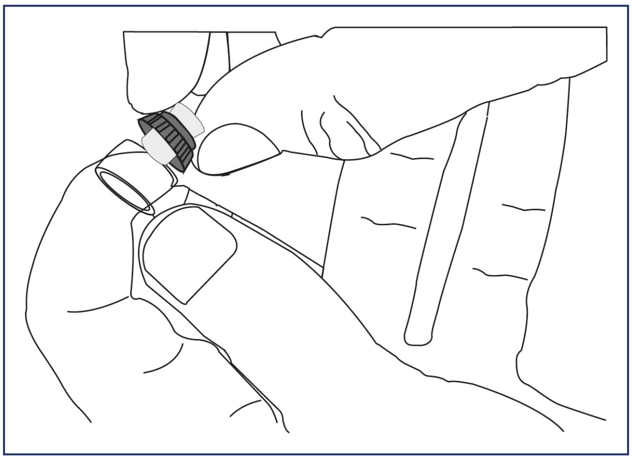
1. Place the sponges in the electrolyte solution and let them soak for the following durations:
   - Approximately 20 minutes in total for previously unused sponges.
     New sponges need to be inserted into the holders after 15 minutes:
     Take the sponges out of the solution with the pair of tweezers and insert them into the holders one-by-one. Then, put them back into the solution for 5 more minutes. Make sure the sponges are centered in the holders, i.e., protruding the same distance on both ends.

   - Approximately 5 minutes in total for all further applications.
2. Insert the sponge holder into the electrode holder on the cap. Ensure that it is secure by turning it to the right while firmly gripping the electrode arm.

   It is important not to over-extend the arms by applying counterpressure when screwing in the sponge holders.

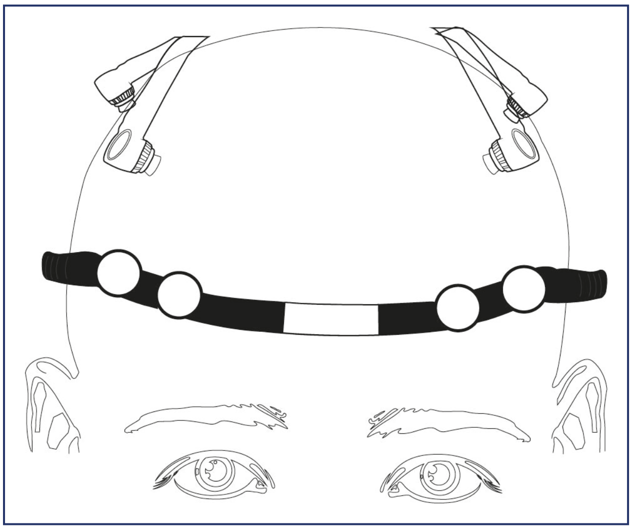
   When inserting the sponge, make sure that no water runs into the headset. This can be ensured by first putting the spongeholders on a towel and removing excess liquid and by holding the X.on in a way that it cannot be hit by potential drops of liquid.



3. Repeat for all electrode holders.
4. Turn on the X.on, refer to How to Turn on the X.on.

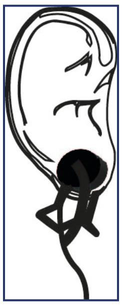
5. Choose the best fitting forehead strap based on head circumference: select the short one for a circumference ‹56 cm and the long one for circumferences ≥56 cm.

   Attach the selected forehead strap to the two connectors on the X.on.

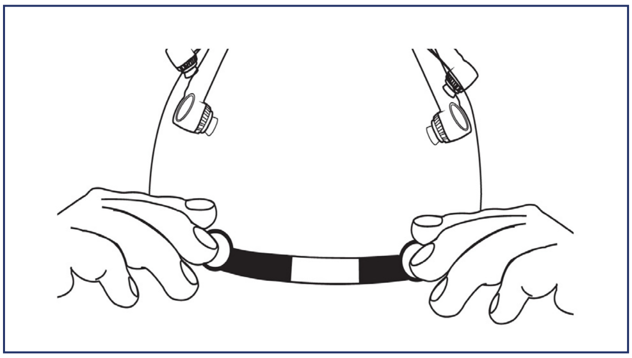
   

6. Place the X.on on the test participant and adjust the electrodes as required.
7. Gently attach the ear clip (GND/REF) to the earlobe on the right ear and connect the cable to the base of the X.on.

   Note: Use the supplied pipette to add a drop of electrolyte solution (from the plastic jar) between both sides of the earlobe and the electrodes on the ear clip (GND/REF).

8. Ensure the X.on is correctly positioned. Adjust the forehead strap as required by pulling the sliders closer together or further apart.

9. Refer to Connect X.on to the X.on App for Android™or Connect X.on to the X.on App for PC.

3.2 How to Turn on the X.on

1. Ensure that the X.on is adequately charged. Refer to How to Recharge the X.on.
2. Locate the power button on the back of the X.on.

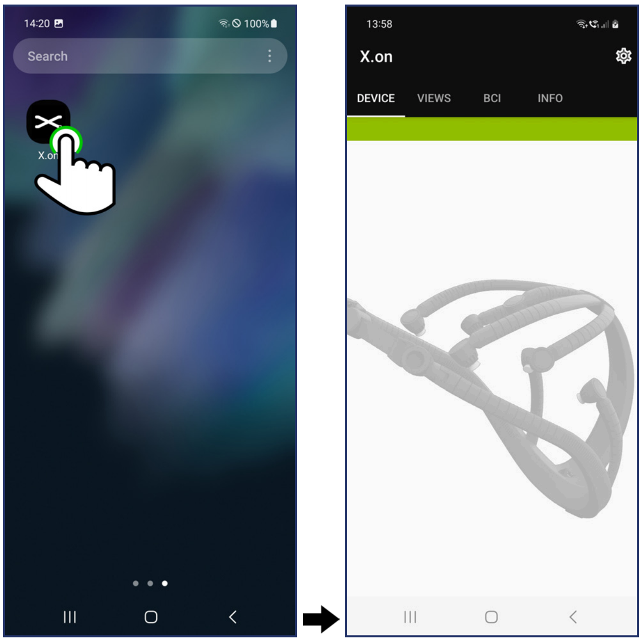
3. Press and hold the power button for 1 second to power on.

The right LED will illuminate green; the left LED will flash blue to show that the X.on is ready to pair with the X.on App for Android™.

3.3 Connect X.on to the X.on App for Android™

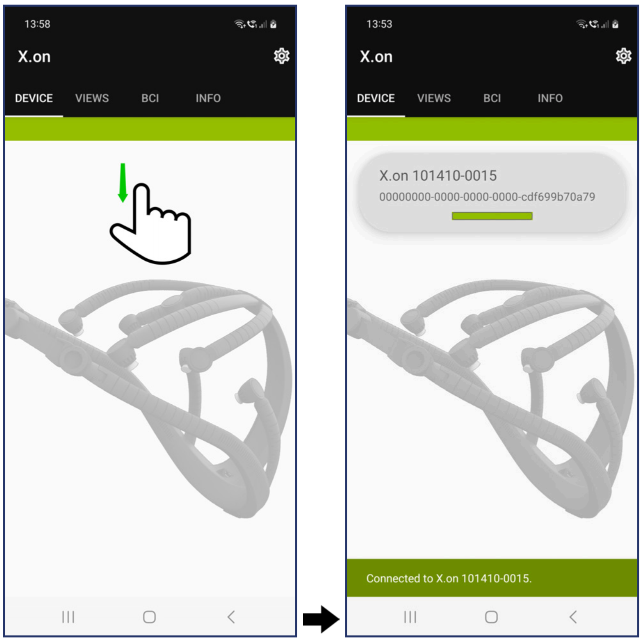
For more information on how to use the X.on App for Android™, refer to section The X.on App for Android™.

1. Turn on Bluetooth® on your Android™ phone.
2. Tap the X.on App for Android™ icon. The app will open on the DEVICE tab.

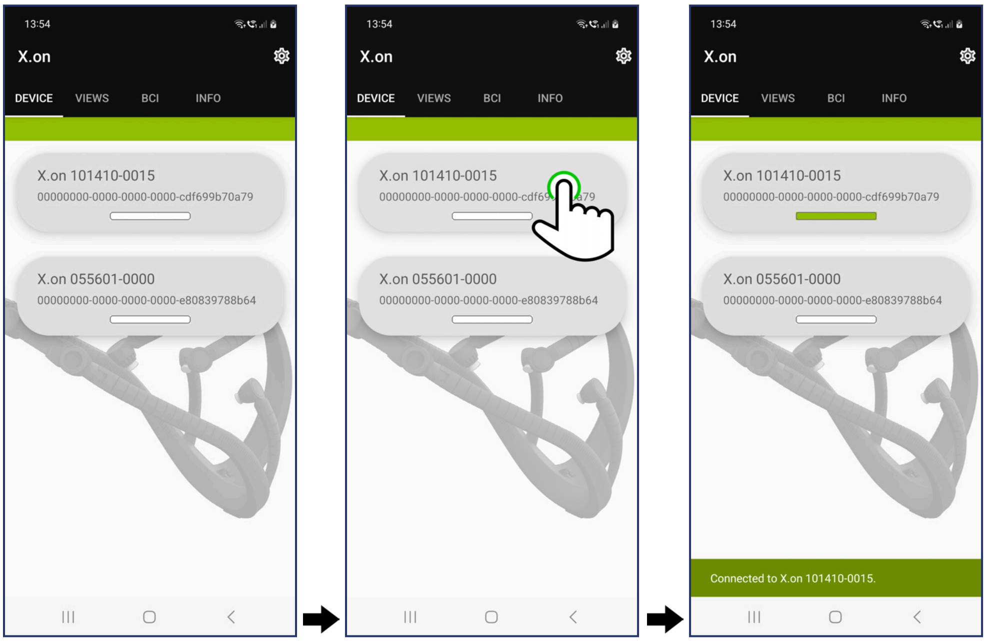


3. Turn on the X.on and check the serial number. Refer to How to Turn on the X.on. The serial number (SN) can be found on the label placed on the inside of the X.on. All the other items on the label are explained in Regulatory Information.

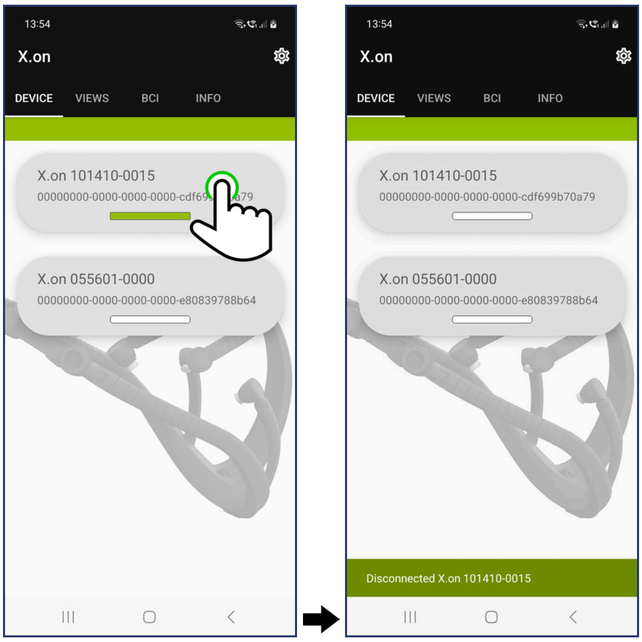
4. Swipe down on the X.on App for Android™ to refresh the DEVICE tab contents. If there is only one X.on in range, the app will automatically connect to the X.on with the corresponding serial number and confirm successful connection. The connected X.on will display with a green bar below its ID consisting of "X.on" and serial number.



5. If multiple X.ons are in range, select the X.on you want to connect to. The chosen X.on will be represented by the green bar.



6. Tap the currently selected X.on to disconnect from it in case you want to connect to another X.on.



3.4 Connect X.on to the X.on App for PC

For more information on how to use the X.on App for PC, refer to section The X.on App for PC.

We recommend using an external Bluetooth® adapter (Bluetooth® version ≥5.0) instead of the built-in device as this typically increases performance of the data transmission.

1. Attach the external Bluetooth® adapter and make sure to deactivate the internal Bluetooth® device.

   Note: Verify that only one Bluetooth® device is activated at the same time in Windows® Device Manager.

2. Turn on Bluetooth® in the Windows® Bluetooth® settings.

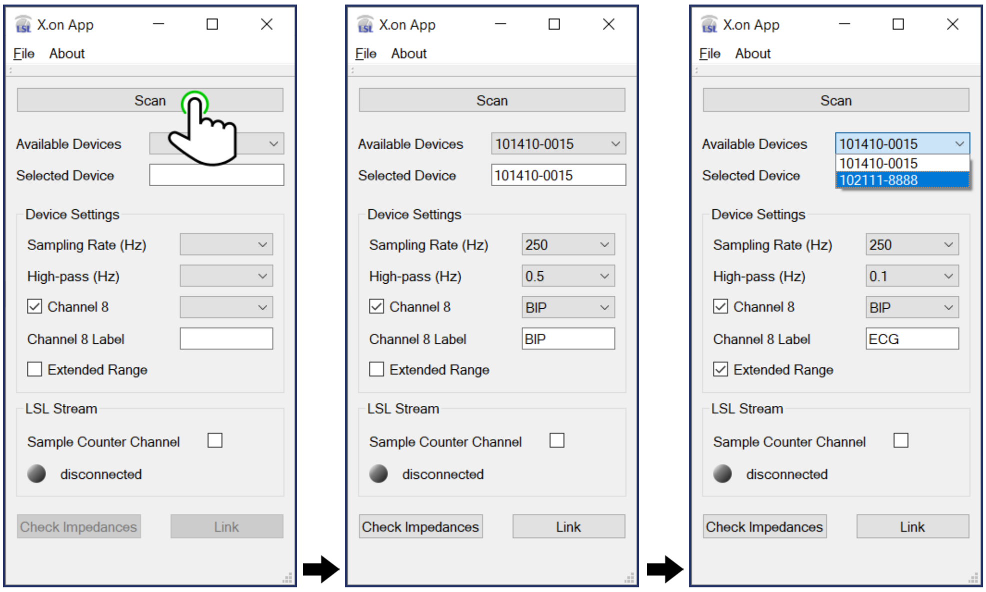
3. Turn on the X.on and check its serial number. Refer to How to Turn on the X.on. The serial number (SN) can be found on the label placed on the inside of the X.on. All the other items on the label are explained in Regulatory Information.

4. In the Windows® Bluetooth settings, add the X.on as an additional Bluetooth device (it will appear as “X.on” plus serial number) and pair it with your Windows® PC.

   Note: Do not forget to unpair when you are done if you want to use your X.on with other devices in the future.

   Note: In Windows® 11, it may be necessary to activate “Advanced” for “Bluetooth devices discovery” if the X.on is not found otherwise.

5. Start the X.on App for PC and click on **Scan**. If there is at least one paired X.on in range, it will be automatically selected. If you want to select a different X.on choose its serial number in the drop-down menu. If the drop-down menu is not greyed out, you can always switch between X.on devices.



3.5 Minimize Impedance Values

It may be necessary to adjust the X.on electrodes to achieve lower impedance values and a higher quality measurement. Impedance values can be monitored in IMPEDANCE mode in the X.on App for Android™ or by clicking **Check Impedances** in the X.on App for PC.

1. Slightly lift the electrode in question from the skin.
2. Move hair aside as this will allow the sponge to have better contact with the skin.
3. Tilt each electrode until it is perpendicular to the head and the surface of the sponge has optimal contact with the skin.
4. Gently apply pressure and move the electrode in small circular motions to further improve contact. Repeat if necessary.
5. Use the supplied pipette to add a few drops of electrolyte solution under the respective electrode, onto the sponge and the skin.
6. It may take a few minutes until the sponges have sufficiently wet the skin.

3.6 How to Turn off the X.on

1. Locate the power button on the X.on.

2. Press and hold the power button for 2 seconds and then release the button to power off the X.on.
   Both LEDs will turn off.

3.7 Cleaning and Handling the X.on

We recommend that you clean the X.on after each use.

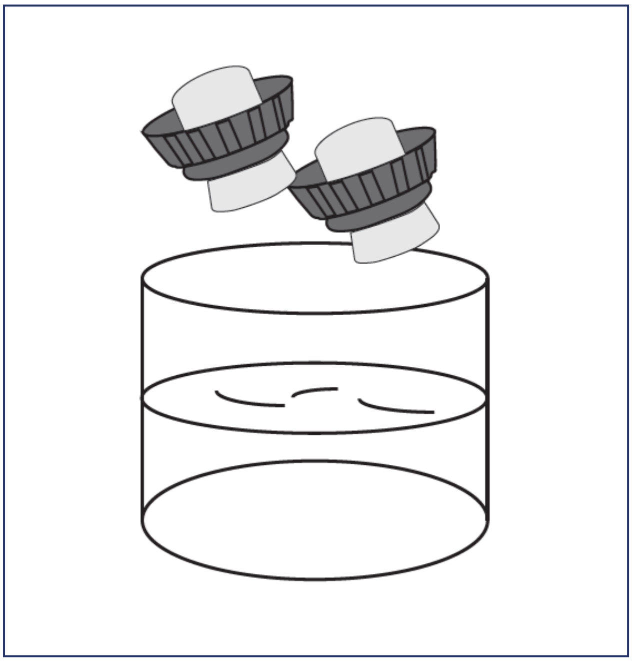
3.7.1 Cleaning the X.on

When your measurements are complete, X.on should be cleaned and prepared for the next participant or storage.

1. Remove the X.on from the participant and turn it off, refer to How to Turn off the X.on.
2. Remove the sponge holders from the electrode holders.

3. Soak the sponge holders in water for a few minutes to clean.

   If you need to replace a sponge, refer to Replacing a Sponge.

4. Store the sponge holders in electrolyte solution until the next use.

   Note: We recommend that the electrolyte solution is replaced every few days. Refer to Preparing the Electrolyte Solution. If you do not plan to use the sponges again within the next 5 days, we recommend to let the sponges dry before putting them in storage.

5. If required, wipe down the X.on with a damp cloth. Before cleaning, disconnect all cables. Connectors must not come into contact with moisture. Moisture causes corrosion.

6. If required, charge the X.on. Refer to How to Recharge the X.on.

3.7.2 Disinfection

Wipe down the surface of the X.on with a cloth drenched in a 70% concentration of isopropyl alcohol (commonly called iso-propanol or 2-propanol) for at least 30 seconds. You can also use the alcohol wipes included in the box.

Please refer to our disinfection guidelines regarding “Electrodes” for the sponge holders and sponges.

3.7.3 How to Recharge the X.on

When the LED displays as red or orange, the X.on should be recharged.

1. Locate the charging port in the rear of the X.on.

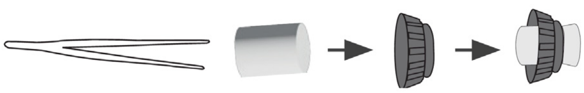
2. Plug the provided USB cable into the X.on charging port and the other end into a USB power adapter or USB port on your computer.
3. The LED will flash green while charging. When charging is complete, the LED will display as solid green.
4. Unplug the USB cable. Your X.on is now ready for use.

3.7.4 Replacing a Sponge

Note: It is easier to change a sponge when it is wet.

Refer to Preparing the X.on for Use.

1. Remove the old wet sponge from the sponge holder and dispose of accordingly.
2. Using the supplied tweezers insert the new wet sponge into the sponge holder.
   Ensure that the sponge is centered in the sponge holder and extrudes evenly from both sides as seen below.



3. Store or use the sponge holder.

3.7.5 Preparing the Electrolyte Solution

Prepare the electrolyte solution as follows:

1. Fill the plastic jar with 80 ml of distilled or de-ionized water. The water should be at least room temperature, but not more than 32 °C (90 °F).
   - Do not use tap water
   - Do not use hot or boiling water
2. Add one spoon (provided with the jar of NaCl) of NaCl to the water.
3. Gently stir the solution until the ingredients are completely dissolved. Dry off the spoon before returning it to the jar of NaCl.
4. (Optional) Add a little baby shampoo (not supplied) and gently stir to dissolve.

4 The X.on App for Android™

Install the provided X.on App for Android™ on an Android™ phone with at least Android™ version 10. Be aware that following screenshots may look different depending on the type of your Android™ phone.

4.1 Download and Install the X.on App for Android™

1. Connect your Android™ phone to the computer via a USB cable.

   Enable file transfer on your phone or make sure to allow the computer to access your phone. The computer will display your Android™ phone under 'Devices and drives'.

2. Using *My Computer* or *Windows® Explorer* navigate to the correct folder and select the file **X.on_App_[VERSION].apk**. Right-click and select **Copy**.

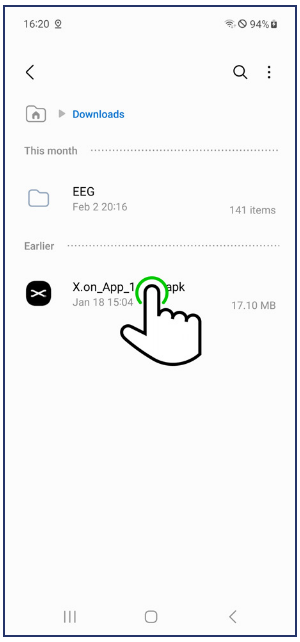
3. Using *My Computer* or *Windows® Explorer* navigate to your Android™ phone and open a folder on the device where you can store the file. Right-click and select **Paste**.

4. Safely disconnect your Android™ phone from the computer.

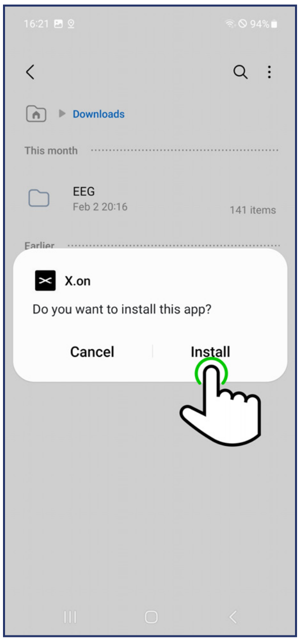
5. On your Android™ phone, navigate to the folder where you stored the file.

   Tap on the **X.on_App_[VERSION].apk** file and tap on **Install** to start the installation process.

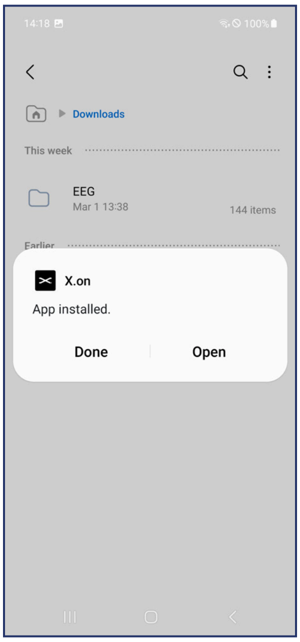
   Please note that the installation procedure may vary depending on your Android™ phone and version.



6. Depending on your phone and Android™ version, acknowledge potential prompts until you can start the installation.



7. The X.on App for Android™ will install.
8. The following (or a similar) screen will display when installation is complete.



9. Select **Done** to finish or **Open** to run the X.on App for Android™.

4.2 Use the X.on App for Android™ for the First Time

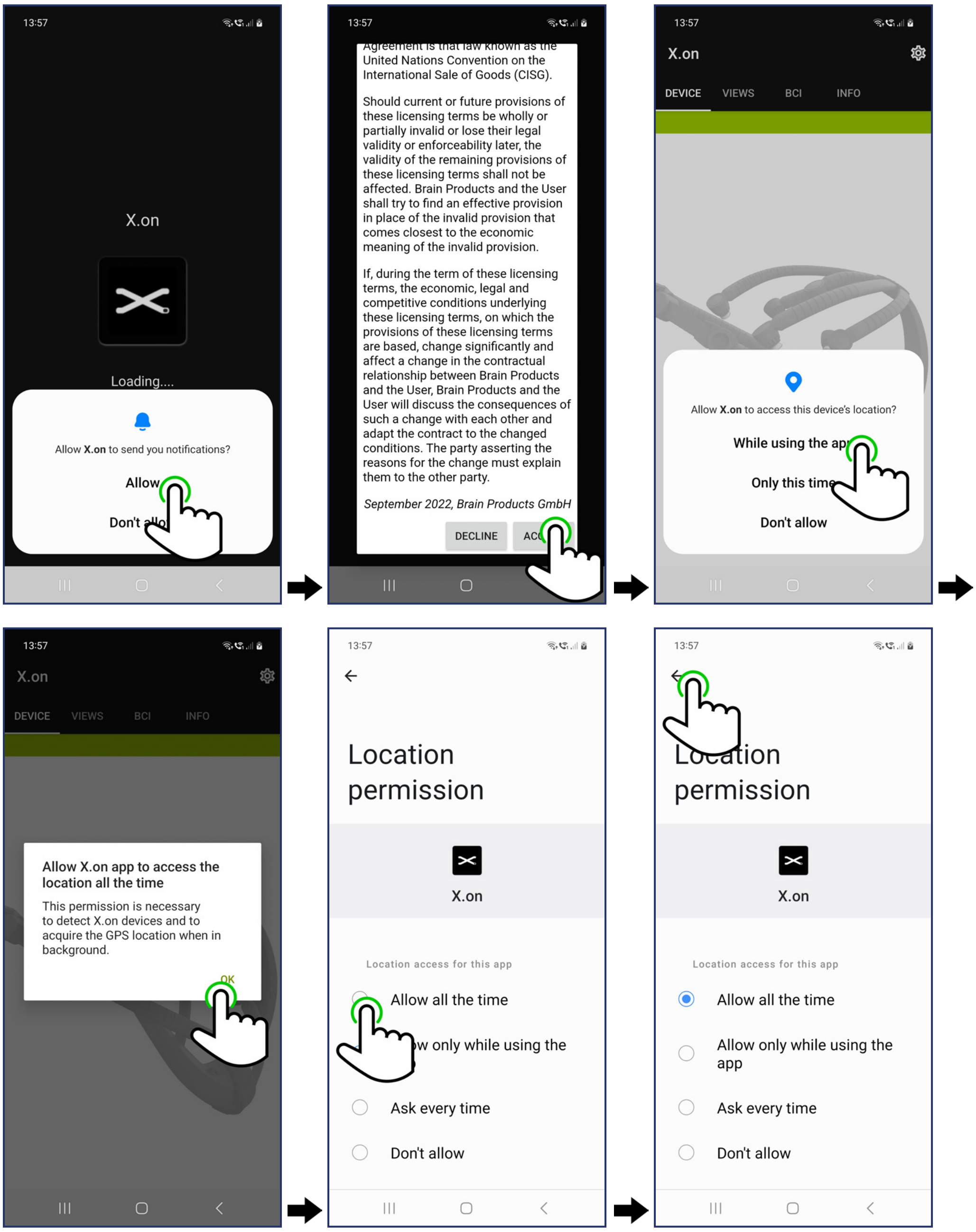
This procedure will only be required when using the X.on App for Android™ for the first time.

1. Tap on the X.on App for Android™ icon.

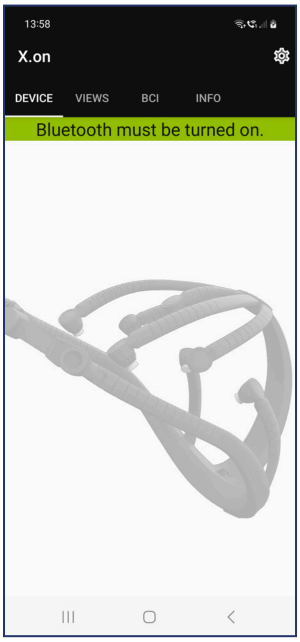
2. For your first use only, you may be required to allow the X.on App for Android™ to access the device's location and allow access to photo, media, and files on your device (it is also possible that you will be prompted at a later time, for example, when setting up the folder for data storage).

   When asked to do so, allow notifications and location services. Allow access to photos, media, and files as well. **Note:** The X.on App for Android™ will only access its own files.

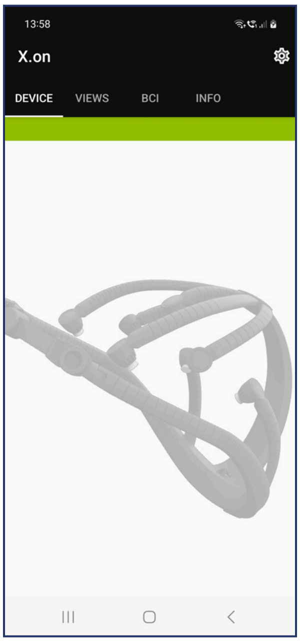
   In addition, before using the X.on App for Android™, you have to acknowledge our license agreement. See the screenshots below for a possible event sequence:



3. The X.on App for Android™ will display the DEVICE tab. You will get a warning if Bluetooth® is turned off.



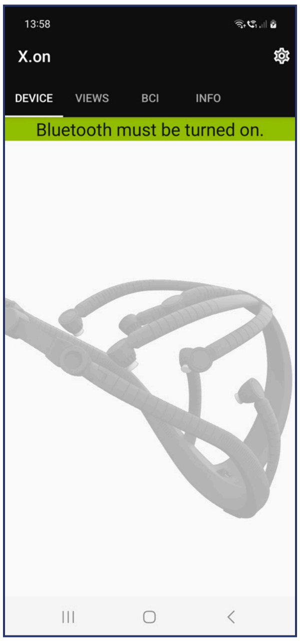
4. Turn on Bluetooth® on your Android™ phone. The warning in the X.on App for Android™ will disappear, The X.on App for Android™ is now ready to pair with your X.on. Refer to Connect X.on to the X.on App for Android™.



4.3 Use the X.on App for Android™ in Demonstration Mode

If you do not have an X.on available it is possible to use the X.on App for Android™ in demonstration mode and have access to all the features.

1. Turn off Bluetooth® on your Android™ phone. When Bluetooth® is turned off, you can use demonstration mode in the X.on App for Android™.
2. Tap the X.on App for Android™ icon. The app will open on the DEVICE tab.

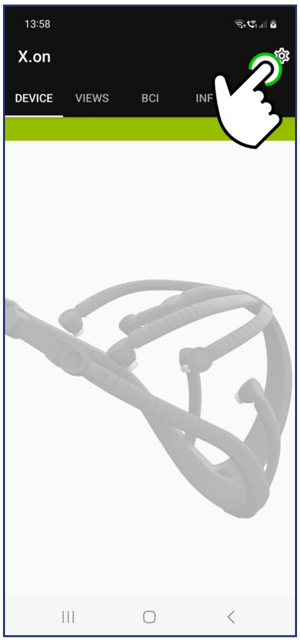


3. Tap on any of the available tabs to navigate the app. Where applicable, simulated signals will be displayed.

4.4 Adjust the X.on App for Android™ Settings

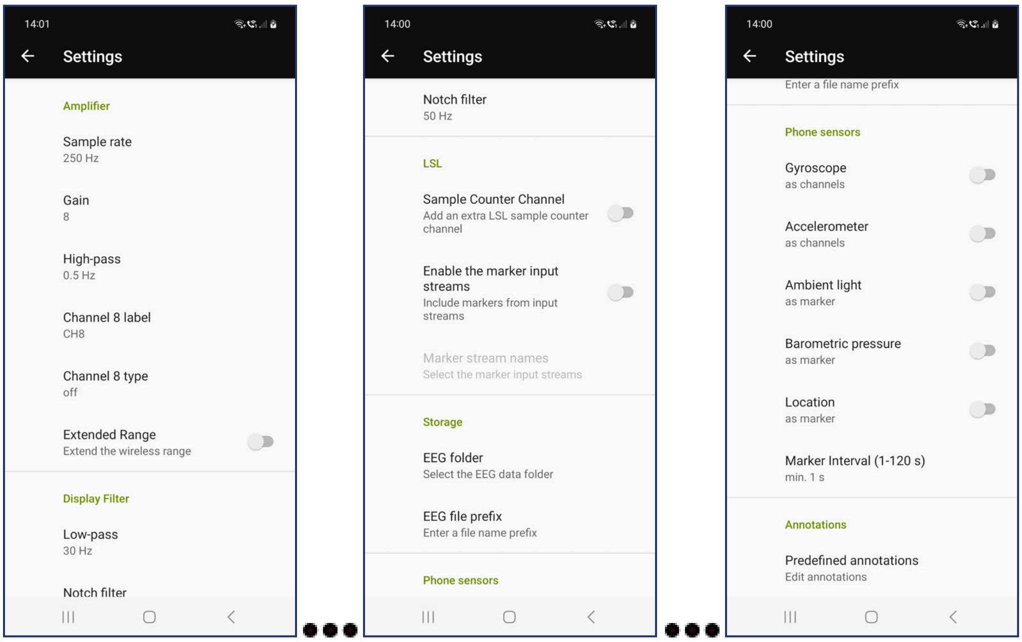
Settings for the X.on App for Android™ can be modified in this section.

1. Tap the X.on App for Android™ icon. The app will open on the DEVICE tab. Tap the Settings icon.



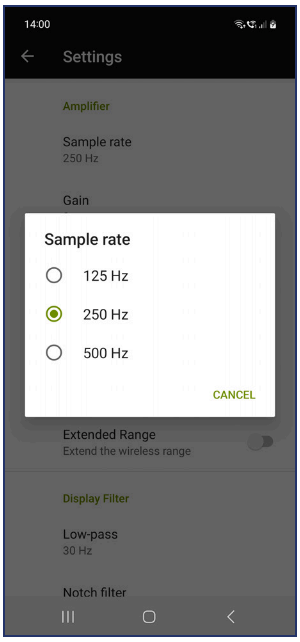
2. The Settings options will display. Tap an option to open adjustable settings or activate/deactivate switches. **Note:** Depending on your phone, the connections status, and previous settings, inaccessible options may be greyed out.

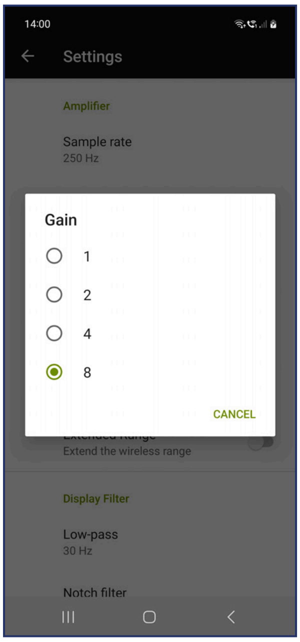
   The following images display the default settings that are applied after a new installation of the X.on App for Android™.

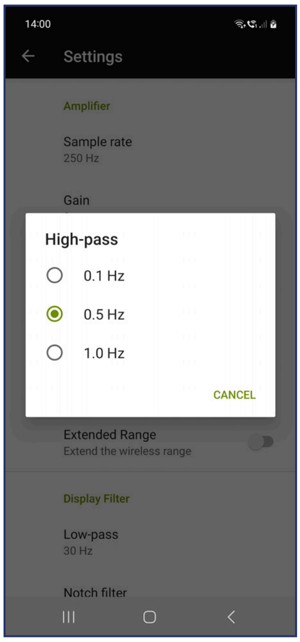


3. In the *Amplifier* section, settings can be changed for **Sample rate**, **Gain** for the analog/digital conversion, and **High-pass**. In addition, the characteristics of the optional **Channel 8** can be set up and the **Extended Range** can be deactivated or activated. **Note:** Activating **Extended Range** limits the **Sample rate** to 250 Hz and the resolution to 16-bit, thereby decreasing the potential of data loss in noisy environments or for measurements that require higher distances between X.on and Android™ phone.

   Note: 500 Hz and 24-bit functionality depends on the brand and type of the used Android™ phone and the integrated Bluetooth® module (at least version 4.2). Please contact info@xon-eeg.com for more information.



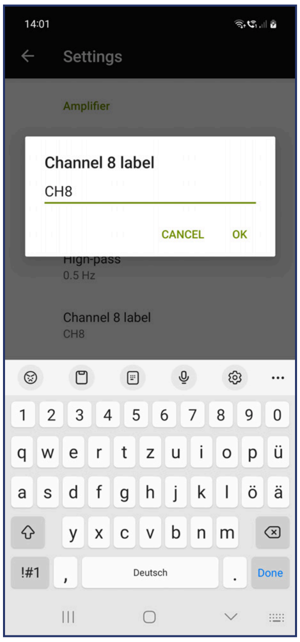


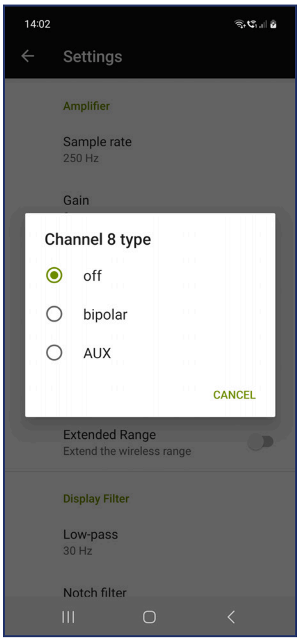


You can choose a name for Channel 8 or choose to deactivate it altogether. If Channel 8 is activated, it can be used as an auxiliary (AUX) signal, or as a bipolar channel. For bipolar recordings, please use the optional accessory "Bip adapter cable", which can be connected to the 3-pin bip/AUX input and allows to use touchproof electrodes.

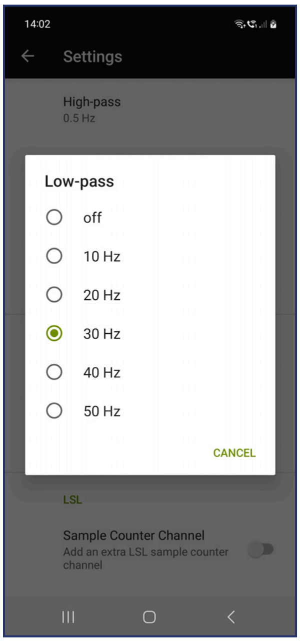
Note: When using the bipolar adapter, make sure to only record physiological signals (e.g., ECG, EMG, EOG).

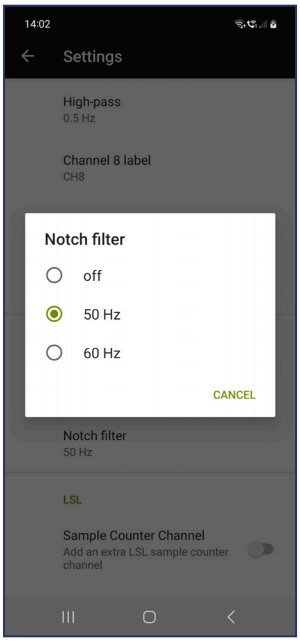
A bipolar channel will use the same filter settings as EEG, whereas AUX automatically sets the gain to 1 and deactivates the high-pass filter.





4. In the *Display Filter* section, settings can be changed for **Low-pass** and **Notch filter**. **Note:** The default settings are displayed below. **Note:** Changing these settings is only relevant for viewing purposes.

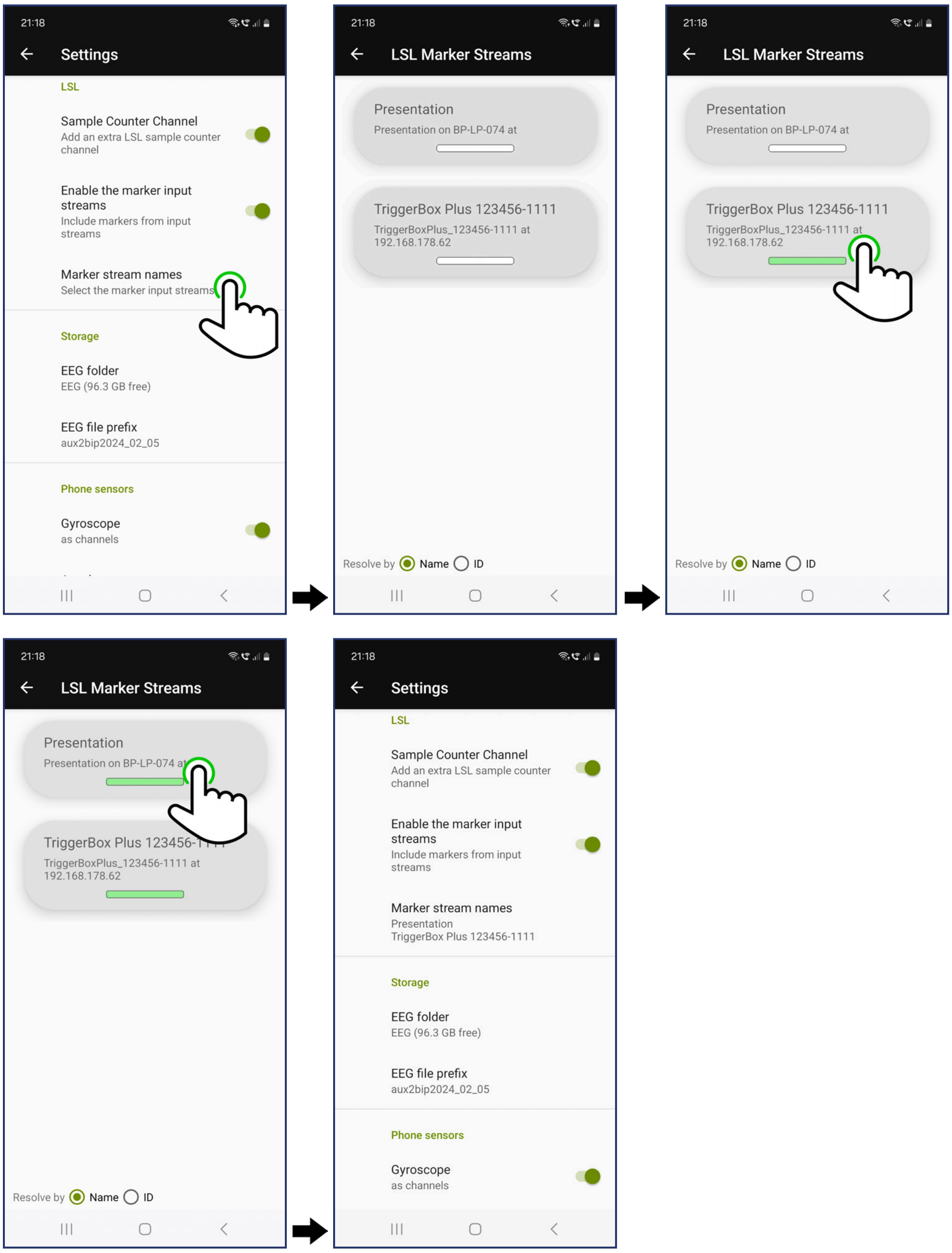




5. In the *LSL* section you can choose to add an extra channel that consists of a counter that increases for every sample, thereby making it easy to spot missing data. If activated, the counter is initialized to zero as soon as MONITOR or RECORD is started.

   If you want to make use of this feature, activate the switch next to **Sample Counter Channel**.

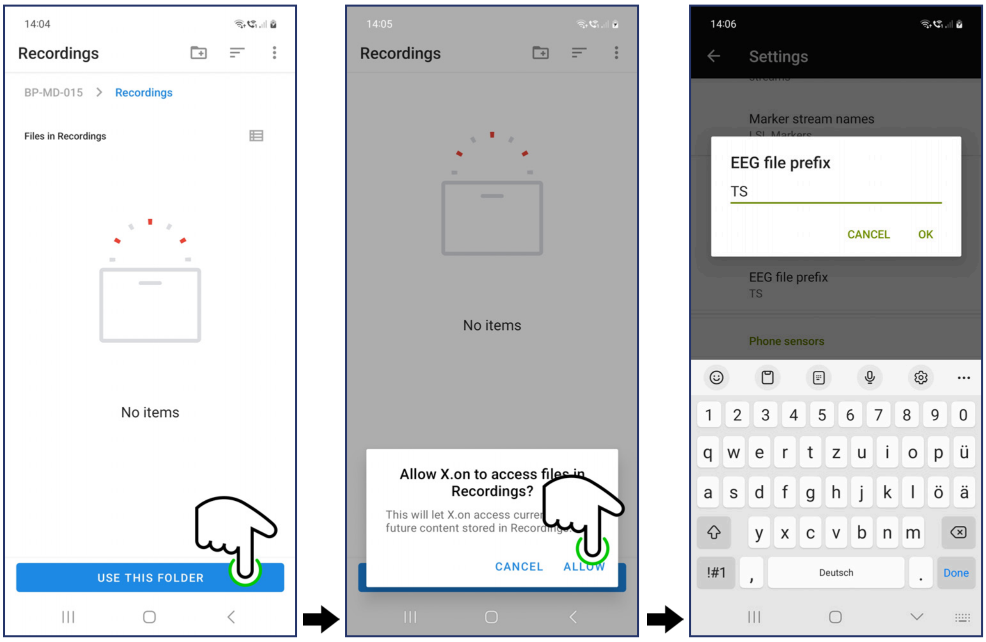
   You can also connect to LSL marker streams that are currently available in the local network. To connect to available streams, tap on **Marker stream names**. All available marker streams will be listed (swiping down refreshes the selection). Tap on each stream to connect to it and tap it again to disconnect. Connected streams are represented with a green bar below their name. When you go back to **Settings**, all selected streams are listed under **Marker stream names**. Markers from selected streams will be displayed, streamed, and saved in your recordings.



6. In the *Storage* section, settings can be changed for the location of the **EEG folder** on your phone and the **EEG file prefix** that is attached to recorded files.

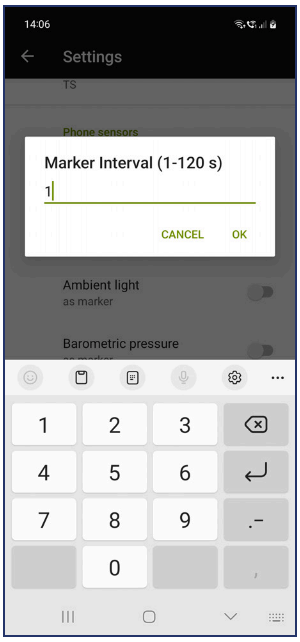
   Note: It may be necessary to specify a folder in order to remove a warning that the **EEG folder** is not accessible, or to allow access to the selected folder. In the example below, a new folder named 'Recordings' is created and selected as the **EEG folder**, and "TS" is selected as **EEG file prefix**.

   Files will be stored in the selected folder in the BrainVision core data format (BVCDF), the names beginning with the selected prefix and ending with an increasing number, as in "TS0001.eeg", "TS0001.vhdr", and "TS0001.vmrk".

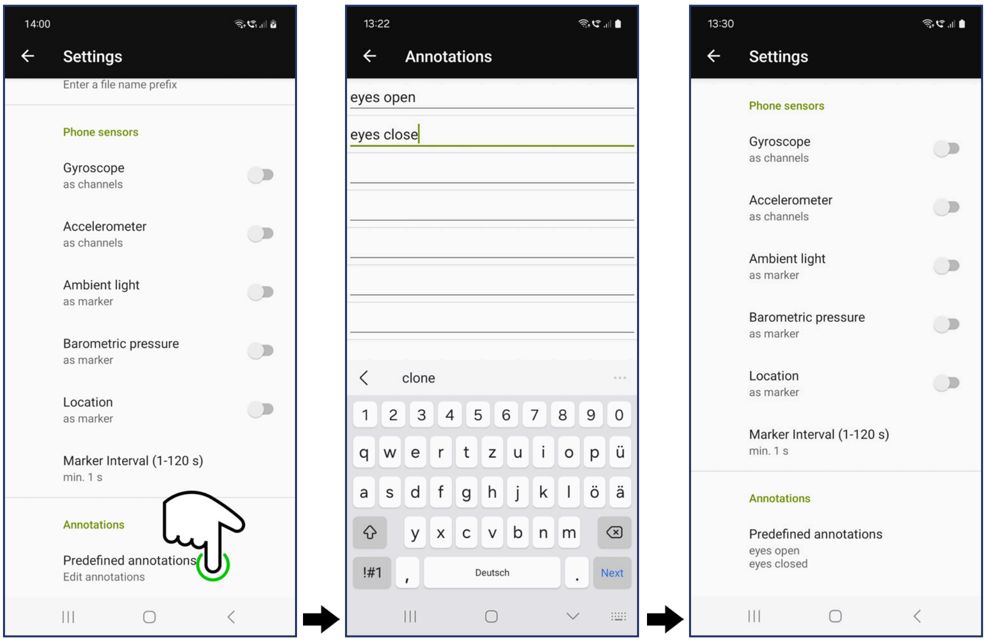


7. In the *Phone sensors* section, you can activate or deactivate sensors that are integrated in your phone. When activated, these sensors are either added as additional signals or markers.

   You can change the minimum time between markers in seconds in the **Marker Interval (1-120 s)**.



8. In the *Annotations* section, you can predefine names for up to ten different annotations. Tap on **Predefined annotations** and then enter names in up to ten rows. These names can later be used as annotations during monitoring or recording your data by double-tapping the screen.



9. Tap the back arrow to return to the DEVICE tab.

4.5 VIEWS Tab

The VIEWS tab provides access to:

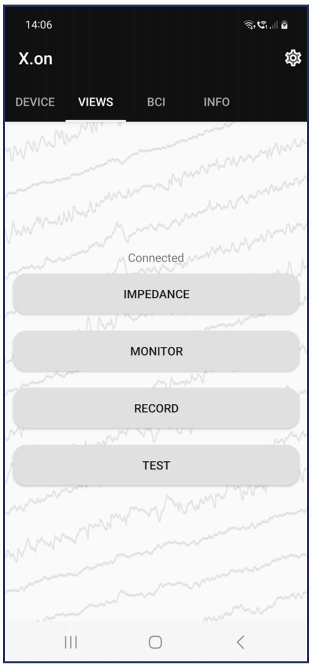
IMPEDANCE

MONITOR

RECORD

TEST

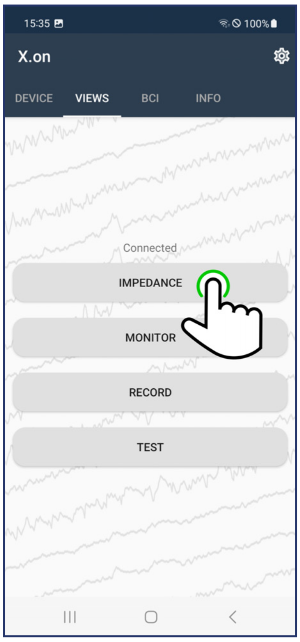
To start the VIEW of choice tap the corresponding option in the VIEWS tab.



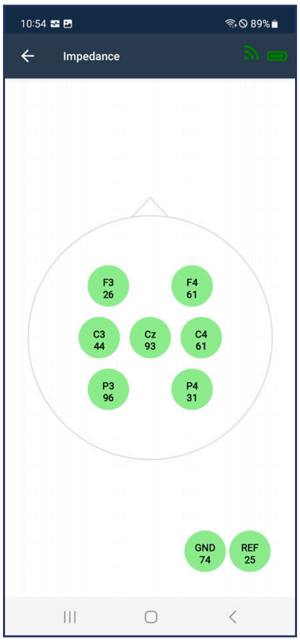
4.5.1 IMPEDANCE

1. Follow the procedure Preparing the X.on for Use.

2. Tap the VIEWS tab to display the options. Tap IMPEDANCE.



3. The X.on electrode positions with corresponding impedance values will display.



Values are shown in kΩ and colors indicate the quality of the contacts.

≤ 100

101–199

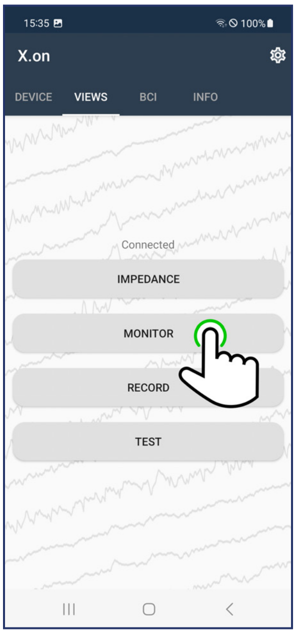
≥ 200

4. To adjust the impedance measurements refer to Minimize Impedance Values.

5. Tap the back arrow to return to the VIEWS tab.

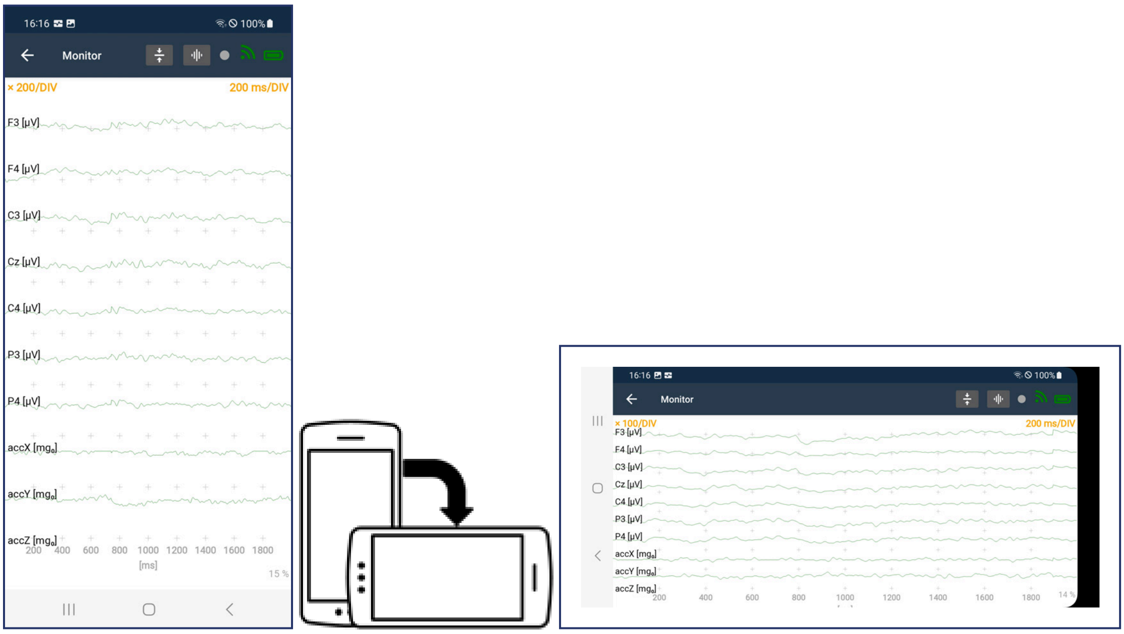
4.5.2 MONITOR

1. Follow the procedure Preparing the X.on for Use.
2. Tap the VIEWS tab to display the options. Tap MONITOR.



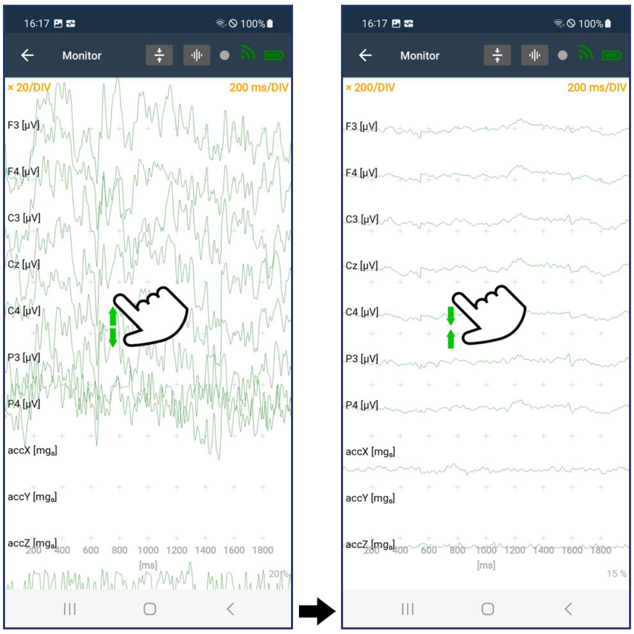
3. You can rotate your phone as required to change the view from portrait to landscape.

   The channels from the X.on and the signals from the Phone sensors (if selected in Settings) are shown on the screen. Bluetooth® transmission quality is represented by the symbol and the battery level by the symbol.



4. To change the amplitude scale, pinch and zoom vertically on the screen.

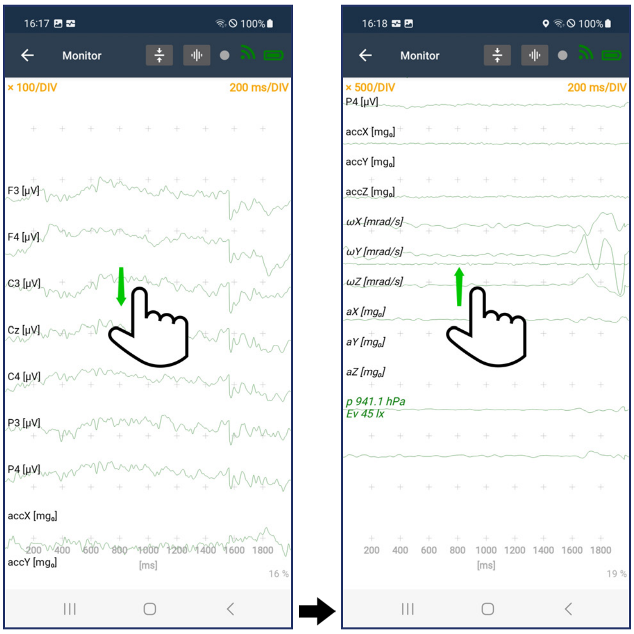
   The amplitude ranges from 500 k/DIV to 1/DIV. The current amplitude scale is shown in the top left of the display and the respective units are shown next to the channel names.



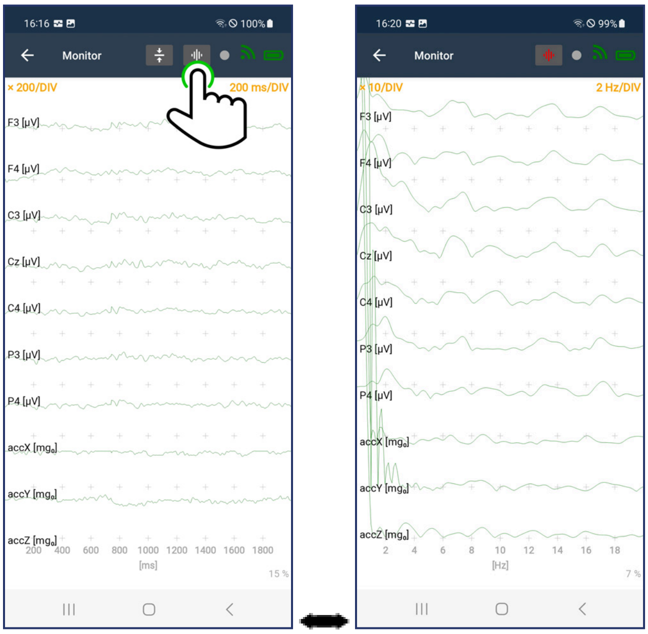
5. To change the time scale, pinch and zoom horizontally on the screen.

   The time scale ranges from 1 s/DIV to 50 ms/DIV. The current time scale is shown in the top right of the display and the labels on the x-axis adapt to the selected range.

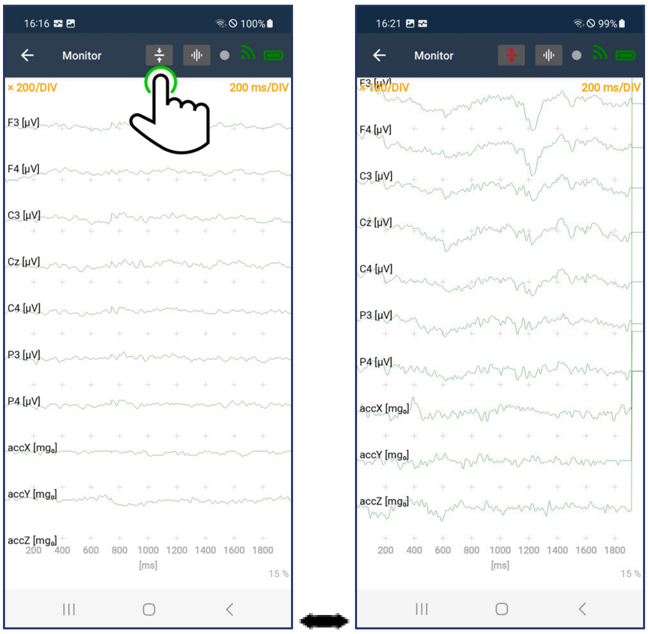
6. Swipe up and down to show channels that do not fit on the screen.



7. Switch between regular time-based signals and Spectrum view, by clicking on the symbol. The color of the symbol changes to red when in Spectrum view. The unit on the x-axis changes from [s] or [ms] to [Hz].



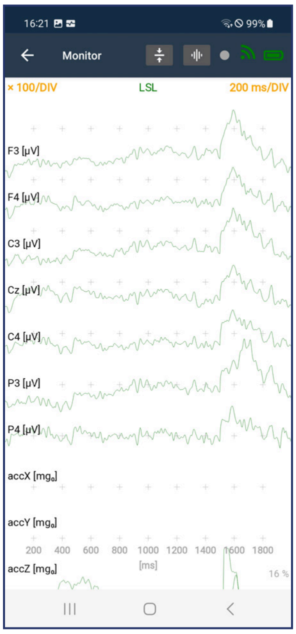
8. Use the visual DC offset correction to display signals with a high offset. If you tap the symbol you can visualize signals that would otherwise be outside of the visible area. This is useful for signals without a high-pass filter, such as AUX or accelerometer signals.



9. The X.on App for Android™ automatically sends LSL streams to the local network when data acquisition is running. If any receiver is connected to these streams, you can see a green LSL symbol on the top of the screen.

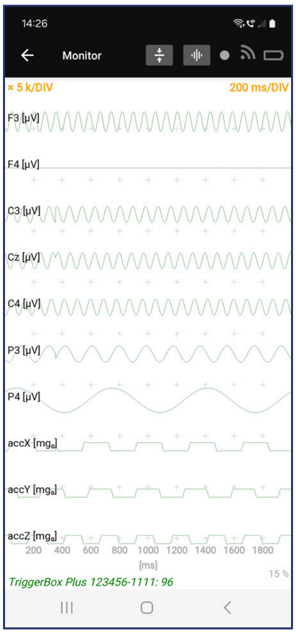
   Note: For best streaming performance make sure to connect your phone to a 5-GHz wireless network.

   Note: Be aware that exiting MONITOR will end LSL streams, no matter whether a client is connected or not.



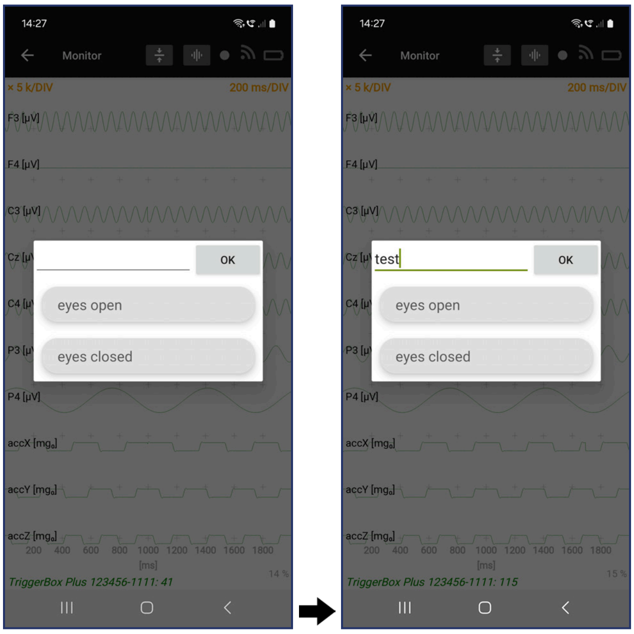
10. If there are active LSL marker streams that were selected in Settings, received markers will be shown on the bottom of the screen (in the example below, markers are received from an LSL marker stream called "TriggerBox Plus 123456-1111").

    Note: The received markers are not only displayed, but are also forwarded in the X.on's own LSL marker stream and/or saved on the phone when recording data in RECORD.



11. It is also possible to add annotations manually. When double-tapping on the screen, a list of predefined annotations is shown, but it is also possible to type new text. The time stamp information is linked to the time of the double-tap, so there is no need to hurry.

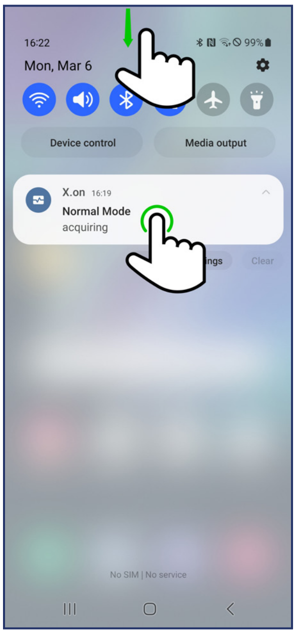
    Note: The annotations are forwarded in the X.on's own marker stream and/or saved on the phone when recording data in RECORD



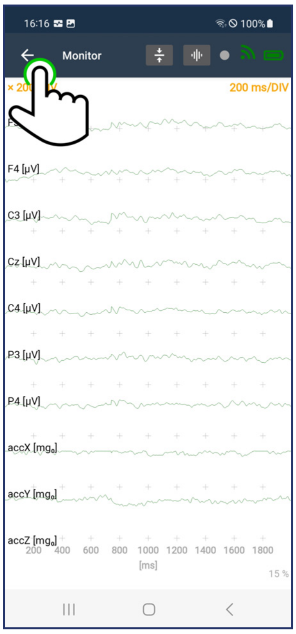
12. Refer to View Selected Channels to highlight specific channels.

13. If the X.on App for Android™ is not active in the foreground, data acquisition will continue to run in the background until stopped.

    Swiping down and selecting the X.on option will open the X.on App for Android™.

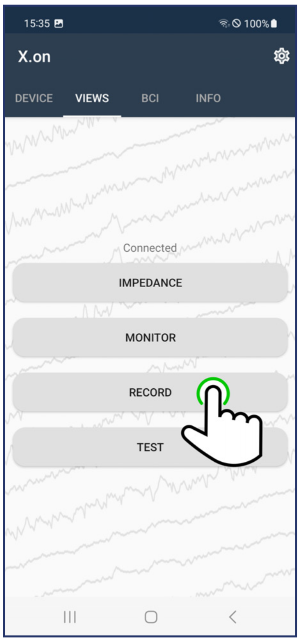


14. Tap the back arrow to return to the VIEWS tab and stop data acquisition.



4.5.3 RECORD

1. Follow the procedure Preparing the X.on for Use.
2. Tap the VIEWS tab to display the options. Tap RECORD.

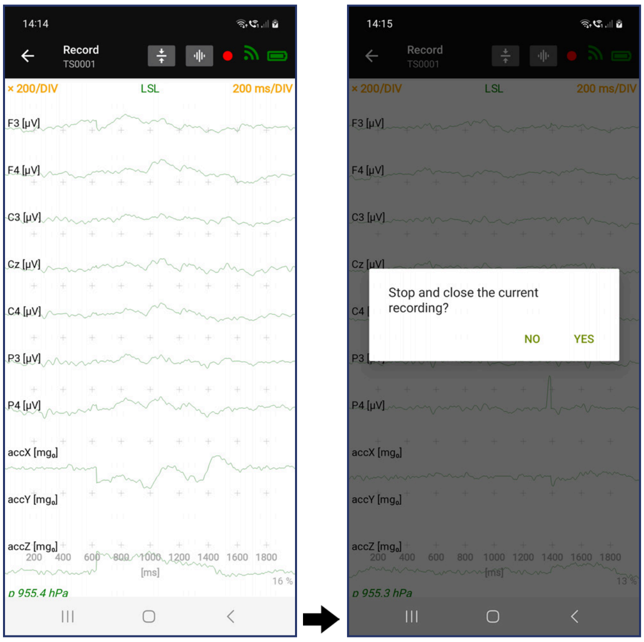


3. RECORD works exactly the same as MONITOR, except that you now save data on your phone. EEG files are recorded in BrainVision core data format (BVCDF) and stored in the folder with the prefix specified in Settings and an increasing number.

   The name of the file is shown on top of the screen and active recording is further indicated by a red circle. After pressing the back icon, you will be prompted with a question to make sure you really want to end the recording. All continuous signals and markers will be saved in the recordings, including markers from external LSL marker streams and manual annotations that were entered via double-tapping.

   Note: It is possible to record to the phone and stream to LSL at the same time.

   Note: Be aware that exiting RECORD will end LSL streams, no matter whether a client is connected or not.

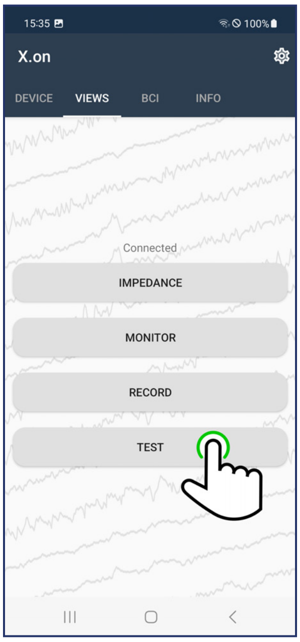


4. Tap the back arrow to return to the VIEWS tab and stop RECORD mode.

4.5.4 TEST

1. Follow the procedure Preparing the X.on for Use.

2. Tap the VIEWS tab to display the options. Tap TEST.



3. TEST works exactly the same as MONITOR, except that you can look at simulated signals instead of real EEG from the X.on. This is to verify a working connection and signal transmission between X.on and phone without requiring full electrode preparation. TEST also requires Bluetooth® to be activated, otherwise signals are only simulated on the phone.

4. Tap the back arrow to return to the VIEWS tab and stop TEST mode.

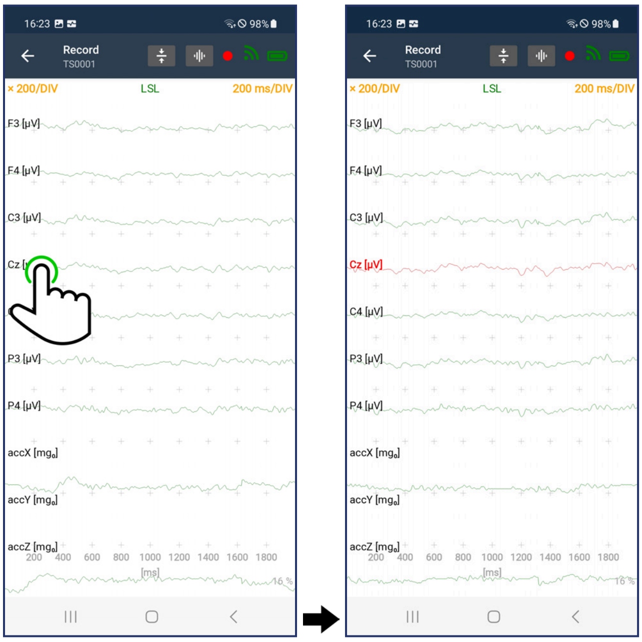
4.5.5 View Selected Channels

1. This option is available when using MONITOR, RECORD, and TEST modes.

2. Use pinch and zoom to zoom in and out as required.
   - To change the time scale, pinch and zoom horizontally on the screen
   - To change the amplitude scale, pinch and zoom vertically on the screen

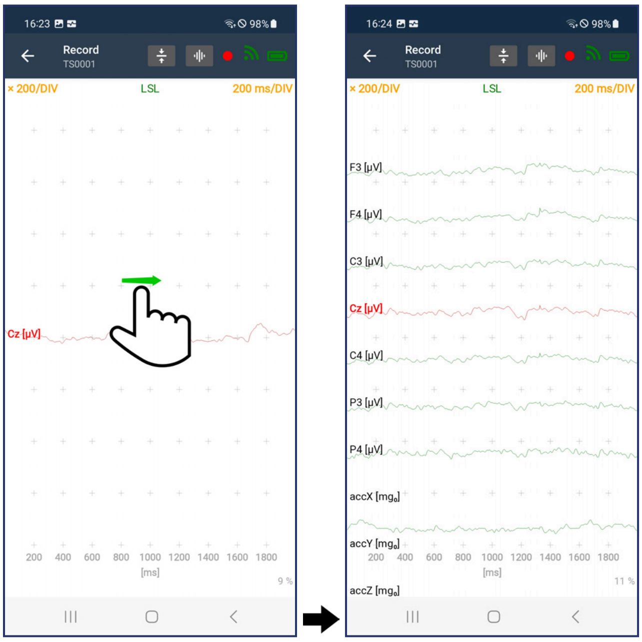
3. While viewing the signals, highlight specific channels by tapping the channel label or the channel signal.

   Note: You can select one or more channels by tapping consecutively. The selected channel(s) and channel label(s) will turn red.

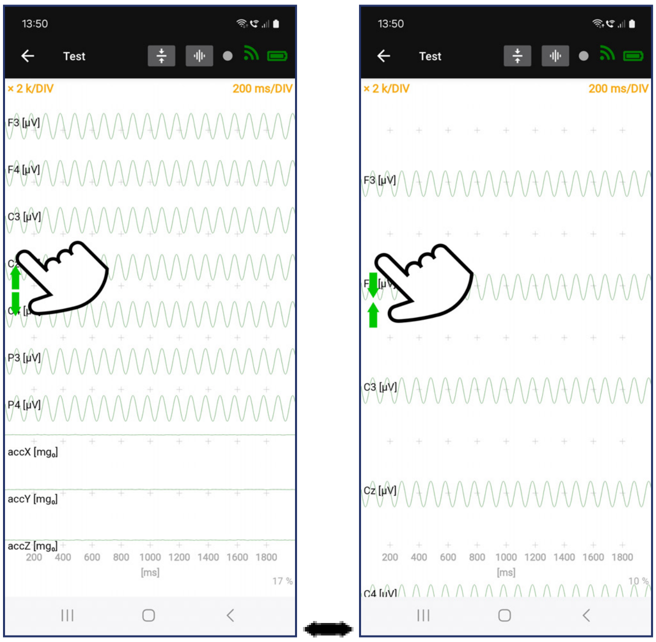


4. To view the selected channel(s) swipe left.

5. To return to the full channel view swipe right.



6. With pinching and zooming vertically over the channel labels on the left side of the screen, you can decrease or increase the number of channels visible.



4.6 BCI (Brain-Computer Interface)

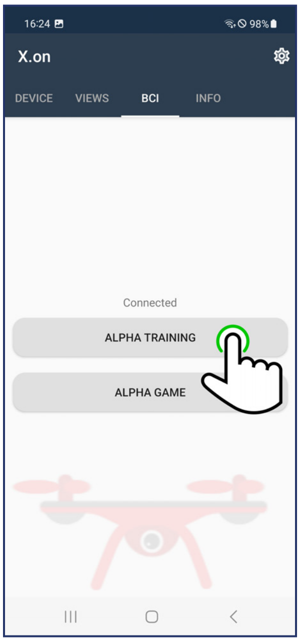
The ALPHA GAME is a demonstration of a brain-computer interface. It uses a classifier based on alpha waves to detect relaxation and let the drone fly.

You must complete ALPHA TRAINING before you can use ALPHA GAME.

4.6.1 ALPHA TRAINING

The training consists of two different types of tasks, a relaxation task and a stress task. ALPHA TRAINING must be completed before using ALPHA GAME.

1. Follow the procedure Preparing the X.on for Use.
2. Tap the BCI tab to display the options. Tap ALPHA TRAINING.

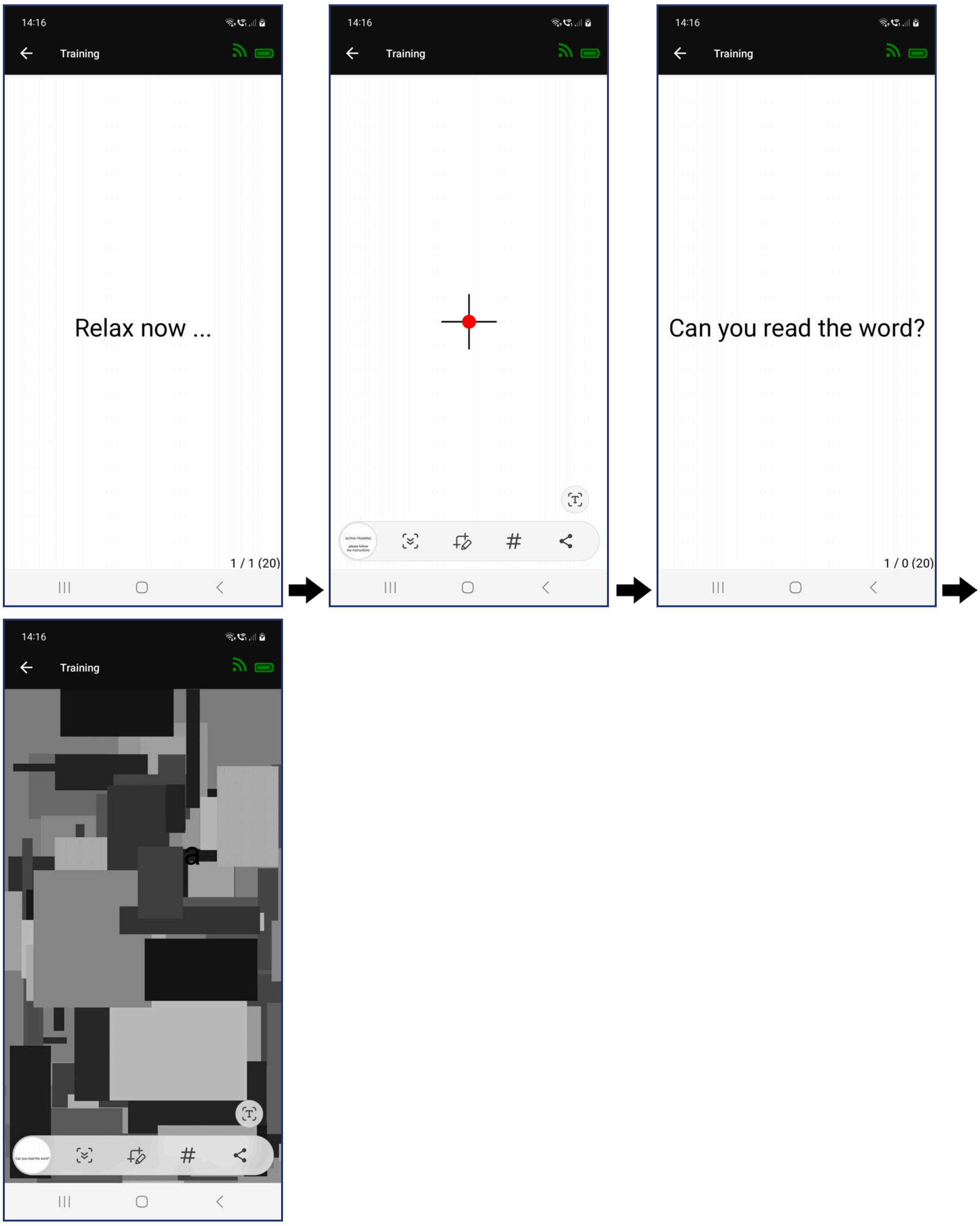


3. The introduction screen to ALPHA TRAINING will display.

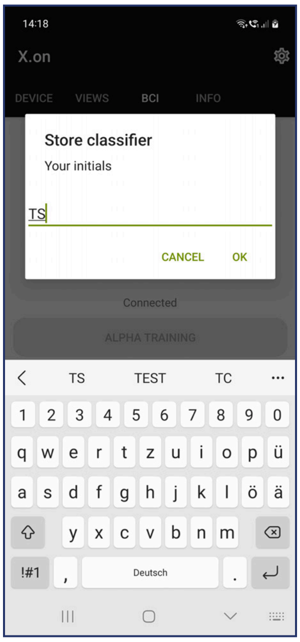
   The counters in the bottom right will increase as you move through the tasks. The first digit represents the relax tasks and the second the stress tasks. The tasks will continue until stopped or until 20 cycles of each are completed.

4. Pay attention to the screen as the trials start.

   Note: We recommend completion of ALPHA TRAINING for better accuracy when using the ALPHA GAME.



5. When you have completed ALPHA TRAINING, the store classifier dialog will display.
   Enter your initials and tap **OK**.
   Note: Your initials (or unique identifier) are required to select your classifier for ALPHA GAME.



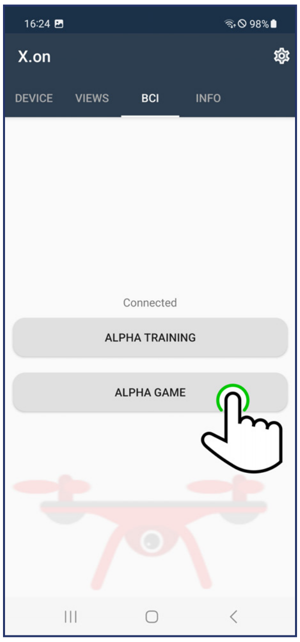
6. You will be returned to the BCI tab.

4.6.2 ALPHA GAME

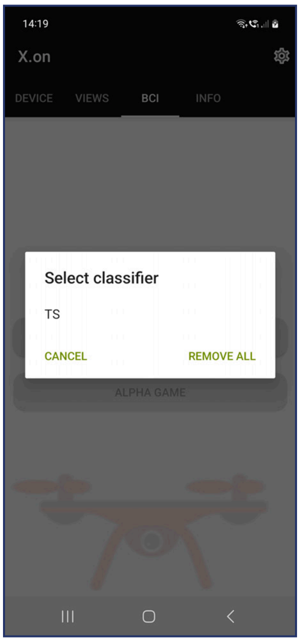
Note: We recommend completion of ALPHA TRAINING for better accuracy when using the ALPHA GAME.

1. Follow the procedure Preparing the X.on for Use.

2. Tap the BCI tab to display the options. Tap ALPHA GAME.



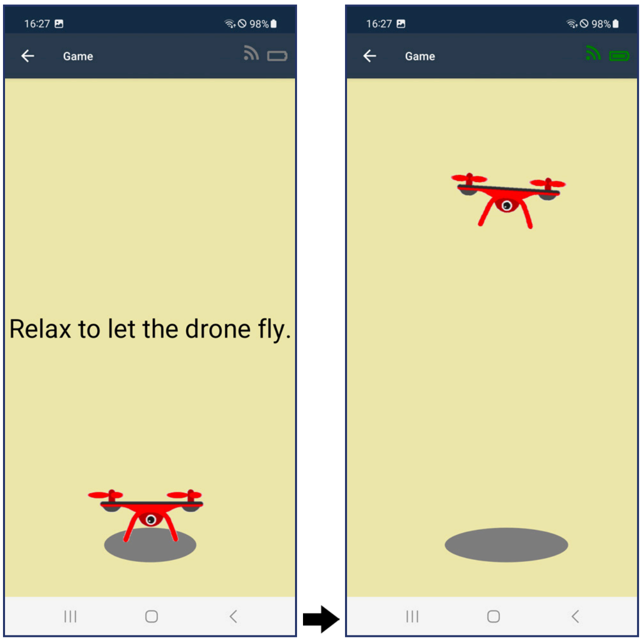
3. Select your classifier (initials) from the list.



Cancel will return you to the BCI tab.

Remove all will ask you to confirm deletion of all classifiers.

4. Now, relax and watch as you control the drone.



5. Tap the back arrow to exit ALPHA GAME and return to the BCI tab.

4.7 View INFO Details

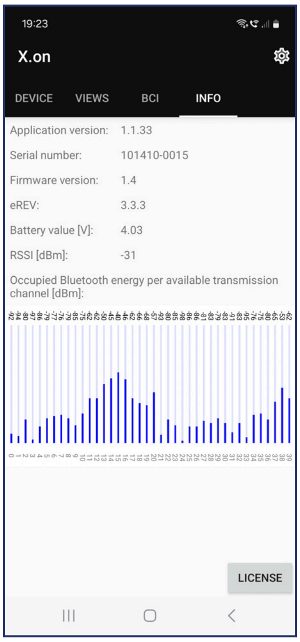
1. Follow the procedure Preparing the X.on for Use.

2. Tap the INFO tab to display the details, such as serial number and software version, but also battery level and signal strength.

   The battery level ranges between bad (‹3300 mV), medium, and good (≥3550 mV). The Bluetooth® connection, or the Received Signal Strength Indicator (RSSI), ranges between bad (≤-90 dBm), medium, and good (›-75 dBm).

   A graphical representation of occupied Bluetooth® energy per available transmission channel in dBm serves as an indicator of the general environmental noise. If you see many blue bars, consider relocating to a different area.

   Tapping on LICENSE gives you the option to review the license agreement.

5 The X.on App for PC

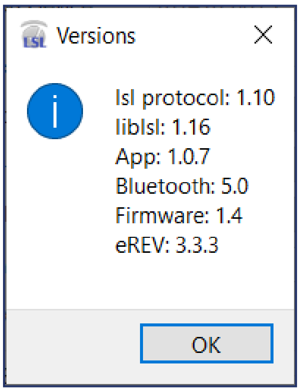
In case you do not want to use a phone, the X.on App for PC is a suitable alternative for Windows® PCs (for Windows® 10 64-bit or Windows® 11 64-bit).

5.1 Download and Install the X.on App for PC

1. Download the file **LSL_Xon-[VERSION]-win64.exe** on your PC and run it.
2. Follow the instructions and agree to the terms of the license agreement, then choose a folder where you want to install the application..
3. Before starting the X.on App for PC, make sure to activate the correct Bluetooth® adapter (see Connect X.on to the X.on App for PC) and make sure to pair your X.on with the PC.

5.2 Start the X.on App for PC

1. Start the X.on App for PC from the *Start* menu (**LSL-X.on**) or from the folder you installed the application in.
2. Connect to the X.on of choice (see Connect X.on to the X.on App for PC).
3. Check information about the PC's and X.on's configuration in the *About/Versions* tab:



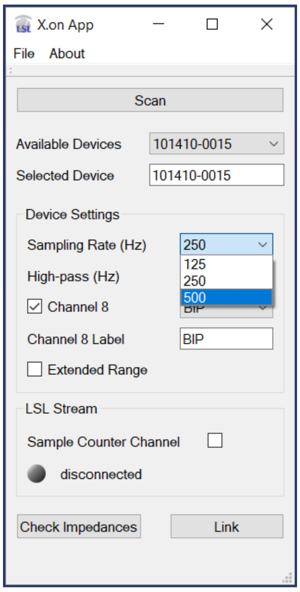
The most important information is the Bluetooth® version, as certain functions (500 Hz and 24-bit transmission) are only possible for Bluetooth® version ≥4.2).

5.3 Adjust the X.on App for PC Settings

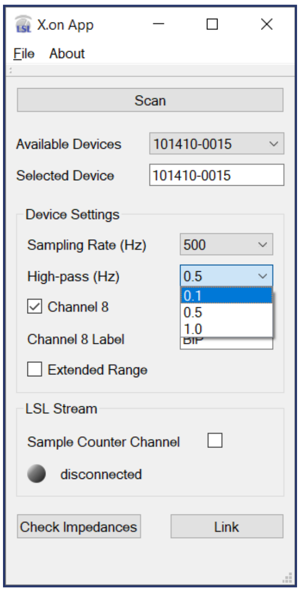
When the X.on is properly connected to theX.on App for PC, you can adjust the settings according to your requirements.

1. The **Sampling Rate (Hz)** can be selected between 125, 250, or 500 Hz.

   Note: 500 Hz can only be selected if the used Bluetooth® version is at least 4.2 and **Extended Range** is not activated.

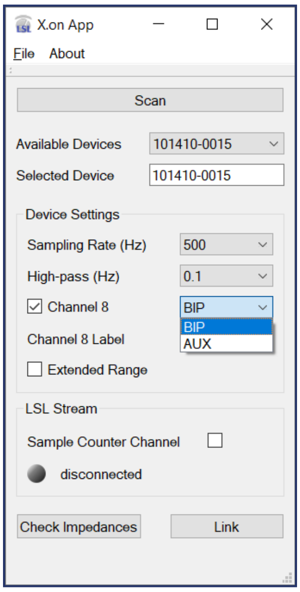
2. The **High-Pass (Hz)** filter can be set to 0.1, 0.5, or 1.0 Hz:

3. You can turn the 8th channel on or off and set it to **AUX** or **BIP** (if the checkbox next to **Channel 8** is activated) and give it a name next to **Channel 8 Label**. If **Channel 8** is activated it can be used as an auxiliary (AUX) signal, or as a bipolar channel. For bipolar recordings, please use the optional accessory "Bip adapter cable", which can be connected to the 3-pin bip/AUX input and allows to use touchproof electrodes.

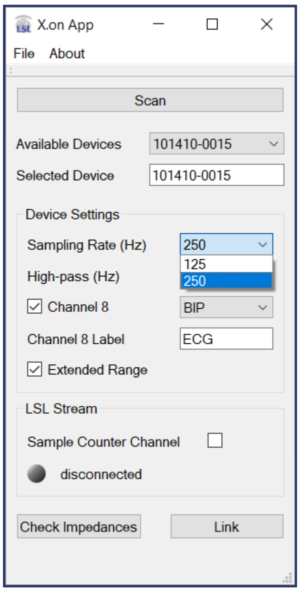
   Note: When using the bipolar adapter, make sure to only record physiological signals (e.g., ECG, EMG, EOG).

   A bipolar channel will use the same filter settings as EEG, whereas AUX automatically sets the gain to 1 and deactivates the high-pass filter.



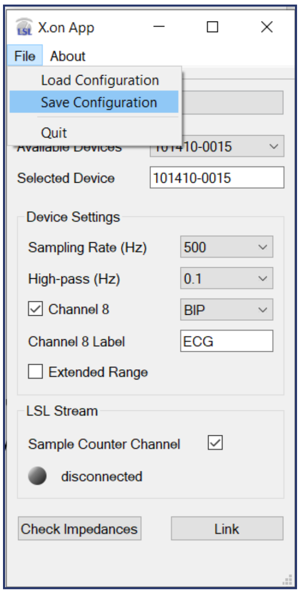
4. If **Extended Range** is activated, the sampling rate is limited to 250 Hz and the resolution to 16-bit. This option decreases the potential of data loss in noisy environments or for measurements that require higher distances between X.on and PC.

   Note: Extended Range is automatically selected for Bluetooth® adapters with a version ‹4.2.



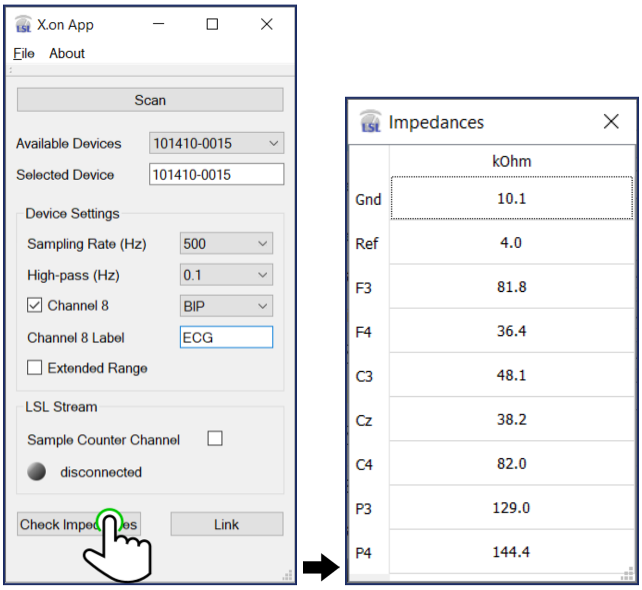
5. In the **LSL Stream** settings, you can activate an additional **Sample Counter Channel** that makes it easier to check for lost samples in your recording.

6. If you are satisfied with your adjustments, you can save your settings in a config-file via **File/Save Configuration** and later load it again via **File/Load Configuration**.



5.4 Check Impedances in the X.on App for PC

1. Click on **Check Impedances** to display the impedance values of all the electrodes.



The values in the table show the values in kΩ for all electrodes, including ground (Gnd) and reference (Ref) electrodes. To adjust the impedance measurements refer to Minimize Impedance Values.

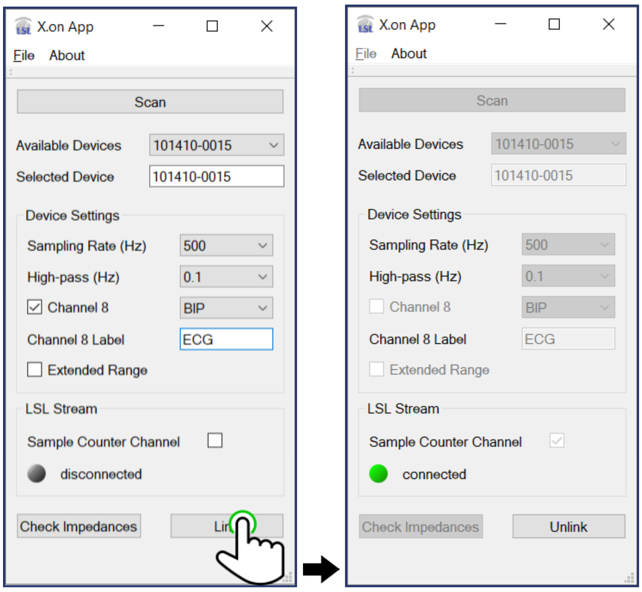
2. When you are satisfied with the impedance values, close the **Impedances** window to return to the X.on App for PC.

5.5 Stream EEG Data via LSL with the X.on App for PC

The X.on App for PC can provide an LSL stream to the local network. Clients for viewing or recording the data can connect to the stream that includes in its name the phrase “X.on”, the serial number, and the sampling rate.

The stream will include 7 EEG channels, optionally 1 auxiliary or bipolar channel, 3 accelerometer channels (in x-, y-, and z-direction), and optionally 1 sample counter channel.

1. Click on **Link** to start the LSL streaming.



   When data is streamed via LSL, the color of the circle turns to green and the text turns from “disconnected” to “connected”.

   Note: if there are problems with your connection, lost samples will be indicated in the X.on App for PC and you will get a warning if the connection between X.on and PC is bad, along with suggestions on how to improve the transmission (e.g., using an external Bluetooth® adapter, reducing the sampling rate, or keeping a closer distance between X.on and PC).

2. When you are done, streaming can be stopped by clicking **Unlink**.

   Note: Make sure to stop recording on any connected client first before stopping the LSL stream.

6 Other Product Information

This section contains the following topics:

Regulatory Information

Maintenance, Repairs, and Disposal

Technical Data

Further information: For frequently asked questions, tutorial videos, and more information on how to get in touch with us, please visit xon-eeg.com.

6.1 Regulatory Information

6.1.1 Manufacturer

Brain Products GmbH
Zeppelinstrasse 7
82205 Gilching
Germany

Tel: +49 (0) 8105 733 84 - 0
Fax: +49 (0) 8105 733 84 - 505

Web: https://xon-eeg.com
Email: info@xon-eeg.com

Brain Products GmbH is registered as a manufacturer of electrical and/or electronic equipment with the stiftung elektro-altgeräte register, Nordostpark 72, 90411 Nürnberg, Germany, under the WEEE registration number DE89975135.

6.1.2 About the product

Markings on the product

	Brain Products GmbH confirms the compatibility of this product according to the Directive 2014/53/EU (RED) of the European Parliament and the Council of 16 April 2014 on the harmonization of the laws of the Member States relating to the making available on the market of radio equipment. Brain Products GmbH confirms the RoHS compliance of this product according to the Directive 2011/65/EU of the European Parliament and the Council of 8 June 2011 on the restriction of the use of certain hazardous substances in electrical and electronic equipment (recast published in the Official Journal of the European Union on 1 July 2011) and its amendment the Commission delegated directive (EU) 2015/863 of 31 March 2015 (published in the Official Journal of the European Union on 4 June 2015).
	Brain Products GmbH confirms the compatibility of this product according to the Radio Equipment Regulations 2017 of 4 December 2017. Brain Products GmbH confirms the RoHS compliance of this product according to The Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012 of 15 November 2016.

	This symbol confirms compliance with the environmental requirements for electronic products.
	Defective products or products which are no longer required must not be disposed of with the household waste. Dispose of them in accordance with national regulations or return the product and its accessories to the manufacturer.
	This device contains lithium-ion batteries. These must be removed before disposal and disposed of separately as a battery.
	Observe the operating instructions.
	MR Unsafe: Products with this symbol must not be used in an MR environment.
	USA Contains FCC ID: XPYNINAB30 This device complies with the USA FCC Federal Communications Commission CFR47 Part 15. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation. This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy, and if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures: ► Reorient or relocate the receiving antenna ► Increase the separation between the equipment and receiver ► Consult your Brain Products' distributor for help.

IC	Canada Contains IC: 8595A-NINAB30 This device complies with the USA FCC Federal Communications Commission CFR47 Part 15. Operation is subject to the following two conditions: 1. This device may not cause harmful interference, and 2. This device must accept any interference received, including interference that may cause undesired operation. Under Industry Canada regulations, this radio transmitter can only operate using an antenna of a type and maximum (or lesser) gain approved for the transmitter by Industry Canada. To reduce potential radio interference to other users, the antenna type and its gain should be chosen in such a way that the equivalent isotropically radiated power (e.i.r.p.) is not more than that is necessary for successful communication. Le manuel d’utilisation des appareils radio exempts de licence doit contenir l’énoncé qui suit, ou l’équivalent, à un endroit bien en vue dans le manuel d’utilisation ou sur l’appareil, ou encore aux deux endroits. Le présent appareil est conforme aux CNR d’Industrie Canada applicables aux appareils radio exempts de licence. L’exploitation est autorisée aux deux conditions suivantes: 1. l’appareil ne doit pas produire de brouillage; 2. l’utilisateur de l’appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d’en compromettre le fonctionnement.” Conformément aux réglementations d’Industry Canada, cet émetteur radio ne peut fonctionner qu’à l’aide d’une antenne dont le type et le gain maximal (ou minimal) ont été approuvés pour cet émetteur par Industry Canada. Pour réduire le 25ecess d’interférences avec d’autres utilisateurs, il faut choisir le type d’antenne et son gain de telle sorte que la puissance isotrope rayonnée équivalente (p.i.r.e) ne soit pas supérieure à celle requise pour obtenir une communication satisfaisante.
R	Japan Contains MIC ID: 204-810006 Japan Radio Equipment Compliance: Japanese Technical Regulation Conformity Certification of Specified Radio Equipment (Ordinance of MPT No.37, 1981), Article 2, Paragraph 1, Item 19, “2.4 GHz band wide band low power data communication system”.
	Next to this symbol, the name and address of the product manufacturer is specified.
SN	SN indicates the unique serial number of the X.on
REF	REF indicates the article number of the X.on

6.2 Maintenance, Repairs, and Disposal

6.2.1 Maintenance

X.on requires no maintenance. Repairs may only be carried out by Brain Products.

6.2.2 Repairs

Before shipping the X.on to Brain Products for repairs, you have to remove the complete battery compartment. To remove the compartment, screw open the lid on the bottom of the X.on and unplug the cable attached to the compartment. The opening has to be left empty for transport. Do not remove the batteries from their compartment.

6.2.3 Disposal

As soon as the product, accessories and cables have reached the end of their service life, dispose of them in accordance with the relevant national regulations. In Germany, for example, the legislation governing electrical and electronic equipment (known as the ElektroG) and the legislation governing batteries (known as BatterieG) are applicable. In the EU and EFTA, the WEEE Directive 2012/19/EU on Waste Electrical and Electronic Equipment as well as the Directive on Batteries and Accumulators and Waste Batteries and Accumulators (Battery Directive) 2006/66/EG apply.

Do not dispose of your product, accessories and cables with ordinary household waste.

Subject to the provision that only original equipment supplied by Brain Products is involved, Brain Products will accept return of the equipment and handle disposal on request.

Data protection: We would like to point out to all end users of the products that you are responsible for deleting personal data on the products to be disposed of.

6.3 Technical Data

Specification	Value
Wireless data transmission	Bluetooth® Low Energy (Version 5). In the 2.4 GHz ISM band. There are no restrictions for its use worldwide.
Electrode technology	Passive, sponge-based
EEG channels	7 (F3, F4, C3, Cz, C4, P3, P4) + ear clip (GND/REF)
AUX channels	1
Size	One size fits all (adolescents through adults) Size range: 52–60 cm head circumference
Weight	~110 g
Amplifier sampling rate	125, 250, or 500 Hz*
A/D conversion	16-bit or 24-bit*
Input voltage range	+/-9.76 mV (16-bit) +/-312 mV (24-bit)
Input impedance	0.412 GΩ
CMRR	›75 dB
Bandwidth	0.1–131 Hz

Battery	Up to 5 h operation charging via USB micro connector Rechargeable lithium-ion button cell Graphite-layered metal oxide ($LiNi_xMN_yCo_zO_2$)
X.on App for Android™	Phone with Android™ v10 or higher required**
X.on App for PC	Windows® 10 64-bit or Windows® 11 64-bit
Data storage	BrainVision core data format (BVCDF, .eeg, .vhdr, .vmkr) on smartphone Extensible data format (XDF) on computer via LabStreamingLayer (LSL)
Raw data access	Offline from phone (BVCDF) or from computer (XDF) Online via LSL
Transportation	The battery in X.on fulfills the requirements of the UN Manual of Tests and Criteria, Part III, subsection 38.3. for transportation
Medical device	No
CE marking	Yes Radio Equipment Directive (RED) 2014/53/EU Directive 2011/65/EU and its amendment Directive (EU) 2015/863
UKCA marking	Yes Radio Equipment Regulations 2017 of 4 December 2017 The Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012 of 15 November 2016

* 500 Hz and 24-bit functionality depends on the brand and type of the used Android™ phone/Windows® PC and the integrated Bluetooth® module (at least version 4.2). Please contact info@xon-eeg.com for more information.

** Functionality cannot be guaranteed for every device with a running Android™ v10 or higher installation.

6.3.1 Ambient conditions

Operation	Temperature: 10 °C to 40 °C (50 °F to 104 °F) Relative humidity: 25 % to 85 %, non-condensing Atmospheric pressure: 700 hPa to 1,050 hPa
Storage and transport	Temperature: 0 °C to 40 °C (32 °F to 104 °F) Relative humidity: 45 % to 85 %, non-condensing Atmospheric pressure: 700 hPa to 1,050 hPa
To prevent mechanical damage during transport, please pack the product in such a way that it is not subject to vibration.	